Arithmetic Expression Geometry of Electromagnetism

Ideal-Point Feynman Histories, Legendre Duality, Contact Tearing, and Projective Radiation

Mingli Yuan
Swarma Research
Beijing 100083, China

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Formalized research note. Mathematical and physical status is stated claim by claim; the note is not a paper of the AEG series.

Provenance and draft notice. This note consolidates five working documents of a research dialogue: `electrodynamics/{费曼图, 勒让德形式, 电动力学讨论-01, 电动力学讨论-02, 电动力学讨论-03}.md`. The dialogue's content, order of argument, and physical narrative are kept; what changed is the form: chat front matter is removed, notation is unified with the AEG manuscripts, and every claim is either (i) traced to a proved statement of Paper 0 or Paper IV, (ii) declared as a dictionary hypothesis, or (iii) computed in place with a reproducible check. The human author directs the programme and retains responsibility; the agent performed the formalization, the verifications, and the drafting. A formalization is not independent review.

Abstract

The AEG manuscripts read an arithmetic expression through the ordered histories that evaluate it: an arithmetic expression space carries an assignment function and a metric, the four operations act as motions, and the order defect of two histories with equal total charges is an exact weighted area in the accumulative commutative space ($\mathcal{A}$). This note asks a single question of that machinery: what happens if the ideal point q of an arithmetic history is read as an asymptotic state, the ordinary point p as a finite-scale dressed state, and the additive and multiplicative charge steps as the emission and absorption of a gauge quantum? The answer is developed section by section, and each step is labelled as a proved AEG statement, a declared dictionary hypothesis, or a computation carried out here.

Five results are obtained inside AEG and need no dictionary at all. First, the family of B_∞-histories joining an ultraviolet starting point to one ordinary point is exactly one fiber $\rho^{-1}(g_{p_0\to p})$ of the operator evaluation map; under the charge projection it fans out into a horizontal segment of the $\mathcal{A}$ at fixed multiplicative charge, while under the Reeb quotient it terminates at distinct heights of one Reeb fiber, the height being the consumed additive charge. Second, the rescaled contact form factorizes as $e^{-M}\alpha = \mathrm{d}(e^{-M}a) - e^{-M}\mathrm{d}A$, so the charge projection and the Reeb quotient are Legendre-conjugate readings of one contact curvature: the order defect appears as a tearing Δa of the assignment fiber in the first and as a Reeb drift $\Delta A = -\mu\lambda\,\mathrm{Area}_g(\Omega)$ in the second. Third, for a pair of bivaluations, uniform motion (constant kernel coordinate b) telescopes to holonomy 1 around every closed loop, whereas an acceleration that moves b produces the four-step cross-ratio holonomy $\Lambda(\square) = (1+a_2b_1)(1+a_1b_2)\big/\big((1+a_1b_1)(1+a_2b_2)\big) \neq 1$, with an exact finite-field witness $\Lambda = 4$ over $\mathbb{F}_7$. Fourth, the incidence-coupled transport $\mathcal{T}(e) \in \mathrm{Aff}(1,K)$ reduces on the Borel sector to the affine cocycle of Paper 0 and retains a

non-telescoping multiplicative component off it. Fifth, the Coulomb radial action is the Reeb holonomy of a closed horizontal loop, and the monodromy condition on the quantized Reeb circle returns the Bohr spectrum $E_n = -m\alpha_e^2/(2\hbar^2 n^2)$.

Five further steps require a declared dictionary, and are stated as hypotheses rather than theorems: the identification of the charged additive step X_{-e} with a physical charge $-e$ and of the reflection $a \mapsto -a$ with charge conjugation; the identification of the depth exponent e^{-M} with the Coulomb kernel ℓ_0/r; the identification of the contact pairing $\omega(D_u, D_v) = \mu\lambda$ with a bilinear interaction energy $\kappa = q_1 q_2$; the identification of a velocity with the tilt of a history; and the identification of a nonvanishing projective holonomy with radiated power. Under those five hypotheses the note recovers the Coulomb sign rule (like charges repel, unlike attract), the Ampère coefficient $(1 - \mathbf{v}_1 \cdot \mathbf{v}_2/c^2)$ of two uniformly moving charges, a no-radiation statement for uniform motion with a radiative obstruction for accelerated motion, the B_∞ versus $B_\infty J B_\infty$ separation of Møller and Bhabha scattering, and the hydrogenic spectrum. No statement of quantum electrodynamics is proved here, and none is claimed.

Keywords: arithmetic expression geometry; contact structure; Legendre duality; torsion–Stokes formula; accumulative commutative space; projective holonomy; bivaluation; idempotent transport; Coulomb law; Ampère force; Bohr–Sommerfeld quantization

中文导读（全篇）

本稿把 `electrodynamics/` 下五份讨论（费曼图.md、勒让德形式.md、电动力学讨论-01/02/03.md）整理成一篇正式的英文文稿，保留原有的物理叙事与论证次序，只做三件事：去掉对话性文字、统一记号（与 AEG 手稿的符号一致）、并把每一条断言分清来源——是 Paper 0/Paper IV 已证的定理、是显式声明的「物理字典假设」、还是本稿重新算过并给出可复核数值的结论。

全篇的主线是一条：**从理想点 q 到普通点 p 的穿线簇**。第二、三节把它翻译成算子历史纤维，并在 $\mathcal{A}$ 与切触空间两个投影下读出「裸荷发散」与「Reeb 纤维作用量残差」；第四节由此得到库仑定律的符号规律；第五、六节引入方向角 θ（匀速）与极点跳动（加速），得到安培系数 $1 - \mathbf{v}_1 \cdot \mathbf{v}_2/c^2$ 与「匀速不辐射、加速才辐射」的代数判据；第七节把该判据写成完整的射影挠率传递联络 $\mathcal{T}(e)$，并在 $\mathbb{F}_7$ 上给出精确见证 $\Lambda = 4$；第八、九节给出 Møller/Bhabha 的 Bruhat 扇区对应与玻尔–索末菲能谱 E_n。

需要读者留意的是**分界线**：第一到第五节里真正属于 AEG 内部、不依赖任何物理解释的数学（算子纤维、勒让德对偶、双重撕裂、交叉比和乐、Borel 退化、$\mathbb{F}_7$ 见证、径向作用量）都是可证的；而「荷步 = 电荷」「$e^{-M} = \ell_0/r$」「$\mu\lambda$ 配对 = 相互作用能」「速度 = 世界线倾角」「和乐不为 1 = 有辐射」这五条是**字典假设**，第十节与各节末尾逐条列出。它们支撑的物理结论（库仑、安培、辐射、束缚态）因此都是**条件性**的：在字典成立时成立，而不是从 AEG 无条件推出。全文所有可算的恒等式都在附录 D 里列了复核脚本，29 项全部通过。

Contents

1 Introduction, Notation, and the Standing Dictionary **6**
- 1.1 The question . . . 6
- 1.2 The AEG statements used, by compiled number . . . 6
- 1.3 Notation and sign conventions . . . 7
- 1.4 The five dictionary hypotheses . . . 8
- 1.5 Roadmap . . . 9

2 Ideal-Point Feynman Histories **10**
- 2.1 Two kinds of ideal point . . . 10
- 2.2 The family of histories is one operator fiber . . . 10
- 2.3 Comb diagrams and charge accounting . . . 11
- 2.4 The image in the accumulative commutative space . . . 12
- 2.5 The image in the contact model: Reeb heights . . . 13
- 2.6 Radix shells . . . 13
- 2.7 Panorama . . . 14

3 The Legendre Form: Charge Projection versus Reeb Quotient **14**
- 3.1 Why the mechanical Legendre transform is a contact statement . . . 14
- 3.2 The arithmetic contact form and its two potentials . . . 15
- 3.3 The dual tearing theorem . . . 16
- 3.4 A worked pair: the kernel word . . . 16
- 3.5 The duality table . . . 17
- 3.6 The open thread: from the affine sector to the projective group . . . 17

4 Electrostatics: Coulomb's Law from Contact Curvature **17**
- 4.1 Stage and dictionary . . . 17
- 4.2 The exchange and the interaction energy . . . 18
- 4.3 Like charges: the two-electron case . . . 19
- 4.4 Opposite charges: the electron–positron case . . . 19
- 4.5 Three geometric readings of the sign rule . . . 20
- 4.6 Status of this section . . . 20

5 Moving Charges: Direction Angle and the Ampère Force **21**
- 5.1 Direction angle and inclined histories . . . 21
- 5.2 The indefinite signature of the frame algebra . . . 22
- 5.3 The contact pairing for two moving charges . . . 22
- 5.4 Uniform motion in ripple geometry: compression without radiation . . . 23

6 Accelerated Charges: Moving Poles and Projective Radiation **24**
- 6.1 Ripple pencils with a moving tangency point . . . 24
- 6.2 Bivaluations and the comoving frame . . . 24
- 6.3 Uniform motion: no radiation . . . 25
- 6.4 Accelerated motion: the pole jumps . . . 25
- 6.5 An exact finite witness . . . 26
- 6.6 The physical dictionary, stated as a dictionary . . . 27
- 6.7 Relation to the AEG manuscripts . . . 27

7 The Projective Torsion Transport **28**
7.1 The problem, as AEG states it . . . 28
7.2 The incidence-coupled transport . . . 28
7.3 The Borel reduction . . . 29
7.4 The curvature of the transport . . . 29
7.5 The finite witness, restated . . . 30
7.6 Where this leaves the open problem . . . 30

8 Dynamical Scattering: Møller and Bhabha Sectors **31**
8.1 Two electrons: both channels inside the affine sector . . . 31
8.2 Electron and positron: the second Bruhat cell . . . 31
8.3 What the correspondence does and does not assert . . . 32

9 Bound States: Closed Loops and Bohr–Sommerfeld Quantization **33**
9.1 Hyperbolic, parabolic, and elliptic histories . . . 33
9.2 The Reeb holonomy of a closed orbit is the action . . . 33
9.3 Monodromy and the Bohr–Sommerfeld condition . . . 34
9.4 How semiclassical is this? . . . 35

10 Dictionary, Status Ledger, and Open Ends **35**
10.1 The dictionary . . . 35
10.2 Status ledger . . . 36
10.3 What is exact and what is not . . . 37
10.4 Open ends . . . 37
10.5 Closing remark . . . 38

A Exact Matrix Computations **38**
A.1 The rank-one projectors . . . 38
A.2 The four-step holonomy . . . 38
A.3 Edge transports over the seven-element field . . . 39
A.4 Exhaustive check over the seven-element field . . . 39

B Curvature Normalization of the Projective Connection **40**
B.1 The curvature of the reductive connection . . . 40
B.2 The density of the transport . . . 40
B.3 Reconciliation . . . 40

C The Coulomb Radial Action **41**
C.1 The radial momentum and the turning points . . . 41
C.2 The half-action . . . 41
C.3 Numerical verification . . . 42
C.4 The spectrum . . . 42

D Register of Verified Identities **42**
D.1 The verification script . . . 43
D.2 Register of verified identities . . . 43
D.3 Corrections made to the working notes . . . 44
D.4 Repairs made to the AEG manuscripts . . . 44
D.5 Manuscript inserts prepared from this note . . . 45

D.6 What is not verified 46

1 Introduction, Notation, and the Standing Dictionary

本节给出全篇的问题、舞台与记号，并把后面要用到的 AEG 结论一次列清（引用一律用 Paper 0 / Paper IV 的**编译编号**，不引用对话里的口头编号）。五个「物理字典假设」在这里集中声明，后面每一节用到哪一条都会指明。若只想看结论，可直接跳第十节的对照表。

1.1 The question

AEG studies an arithmetic expression through the ordered histories that evaluate it. Paper 0 builds three carriers for one history word: the basic hyperbolic arithmetic expression space $\mathfrak{E}_0 = (\mathbb{H}^2, g_{\mu,\lambda}, a)$, in which addition translates and multiplication scales ([1], Proposition 3.6 and the table in Subsection 3.4); the accumulative commutative space $\mathcal{A} = \mathbb{R}^2_{(A,M)}$, which records the total additive charge A and multiplicative charge M and forgets their order (Definition 8.1); and the three-dimensional contact model $\mathcal{C} = \mathbb{R}^3_{(u,v,a)}$ with

$$\alpha = \mathrm{d}a - \mu\,\mathrm{d}u - \lambda a\,\mathrm{d}v, \qquad \mathrm{d}\alpha = -\lambda\,\mathrm{d}a \wedge \mathrm{d}v,$$

whose horizontal fields $D_u = \partial_u + \mu\partial_a$ and $D_v = \partial_v + \lambda a\partial_a$ satisfy $[D_u, D_v] = \mu\lambda\,\partial_a$ (Proposition 9.1, Theorem 9.2). The order defect of two histories with equal total charges is an exact weighted area in $\mathcal{A}$ (Theorem 8.9); in $\mathcal{C}$ the same defect has the constant density $\mu\lambda$ (Theorem 9.2 and Proposition 9.6); and after passing to the Reeb quotient $Q(u, v, a) = (-ae^{-\lambda v}, e^{-\lambda v})$ of Proposition 9.8, the defect acquires a second, conjugate reading as a drift along the Reeb fiber (Theorem 9.9).

This note asks what that machinery says when the two ends of a history are given a physical reading. Fix an ideal boundary point $q \in \partial_\infty\mathbb{H}^2$ and an ordinary point $p \in \mathbb{H}^2$. In AEG terms nothing is new: the histories joining a starting point deep in the ultraviolet to p inside B_∞ form the fiber of the operator evaluation map over one affine operator (Proposition 2.3). In physical terms the same object has a standard name. The ideal point is an asymptotic state, the ordinary point is a finite-scale dressed state, the multiplicative steps are scale evolution, and the additive steps are vertex insertions. The question is then simply whether the AEG invariants attached to that family — the fan-out in $\mathcal{A}$, the Reeb drift, the cross-ratio holonomy, the radial action — reproduce the electromagnetic phenomena that the corresponding Feynman histories describe.

1.2 The AEG statements used, by compiled number

All citations below are to the compiled numbering of the AEG manuscripts [1, 2]. The working dialogue from which this note is drawn cited several of them loosely; where the loose citation named a different statement than the one actually used, the corrected reference is given here and the fact is recorded in Appendix D.

reference	statement used here
Paper 0, Prop. 3.6	the grid motions act on the assignment by $a \circ \mathsf{X}_s = a + s$, $a \circ \mathsf{Y}_k = ka$.
Paper 0, Subsec. 3.4	the five-row table of motions, including the reflection $(x, y) \mapsto (-x, y)$ with $a \mapsto -a$.
Paper 0, Thm. 3.9	continuous arithmetic motion: $\frac{\mathrm{d}}{\mathrm{d}s}(a \circ \gamma) = \mu\cos\theta + \lambda(a \circ \gamma)\sin\theta$.
Paper 0, Thm. 4.2, Subsec. 4.3	ripple pencils and ripple histories; the common tangency point may move with the prefix.

continued

reference	statement used here
Paper 0, Prop. 5.4, Subsec. 5.5	fixed points of $z \mapsto a + b/z$ solve $z^2 - az - b = 0$; general Möbius fixed points are eigenlines.
Paper 0, Props. 7.1, 7.4	affine cocycle and the left/right Maurer–Cartan translation forms.
Paper 0, Def. 8.1, Prop. 8.2	the $\mathcal{A}$ and its weighted evaluation $\hat{\xi} = \int_C e^{-M}\mathrm{d}A$.
Paper 0, Thm. 8.6	only the multiplicative charge descends; the kernel word $h(p,q) = \mathsf{M}_q\mathsf{A}_p\mathsf{M}_{-q}\mathsf{A}_{-e^{-q}p}$ has charges $\big(p(1-e^{-q}), 0\big)$.
Paper 0, Thm. 8.9	weighted torsion–Stokes: $\tau(\gamma,\delta) = \iint_\Sigma e^{M_*-M}\mathrm{d}A \wedge \mathrm{d}M$.
Paper 0, Props. 9.1, 9.6, 9.8, Thm. 9.2	the contact form, its curvature $\mu\lambda$, the Reeb quotient, and $d\alpha = -\mu\lambda Q^*\mathrm{dvol}$.
Paper 0, Thm. 9.9	dual tearing and Reeb drift: Δa at fixed charges versus $\Delta A = -\mu\lambda\,\mathrm{Area}_g(\Omega)$ at fixed state.
Paper 0, App. D.1–D.2, D.4	the oriented rectangle with pulled-back form $pq\,\mathrm{d}s \wedge \mathrm{d}t$; the closed loop $\ell = (\mathsf{A}_p, \mathsf{M}_q, \mathsf{A}_{-p}, \mathsf{M}_{-q})$ with $\tau = p(1-e^{-q})$; Darboux coordinates and $R = -(1/\mu)\partial_u$.
Paper IV, Def. 3.1, Thms. 3.2, 3.3, 3.5	regular bivaluations $\mathrm{Biv}_K^{\mathrm{reg}} = \mathbb{P}^1 \times \mathbb{P}^1 \setminus \Delta$; the bivaluation–projector bijection $\Pi_{a,b} = \frac{1}{1+ab} v_a \otimes \varphi_b$; $G/H \cong \mathrm{Biv}_K^{\mathrm{reg}}$; the frame torsor.
Paper IV, Prop. 3.7, Ex. 3.8	finite-field counts $\lvert\mathrm{PGL}_2(\mathbb{F}_q)\rvert = q(q^2-1)$, $\lvert\mathrm{Biv}\rvert = q(q+1)$, $\lvert H\rvert = q-1$; for $q = 7$: 336, 56, 6.
Paper IV, Prop. 4.9	edge labels given only by ratios of frame lifts telescope along every closed path.
Paper IV, Def. 4.10, Thm. 4.11, Rem. 4.12, Ex. 4.13	the incidence-coupled transport $\mathcal{T}(\tilde{e}) \in \mathrm{Aff}(1,K)$, its Borel reduction, the cross-ratio holonomy, the infinitesimal reductive connection, and the $\mathbb{F}_7$ witness $\Lambda = 4$.
Paper IV, Prop. 10.1, Thm. 10.3	Horner histories $\xi = \sum_i d_i k^{n-i}$ and their histogram counts.

1.3 Notation and sign conventions

The note keeps the AEG conventions unchanged; a reader coming from the manuscripts needs only the following table.

symbol	meaning
$\mathfrak{E}_0 = (\mathbb{H}^2, g_{\mu,\lambda}, a)$	basic hyperbolic arithmetic expression space; $a(x,y) = -x/y$
u, v	arithmetic direction coordinates; $A = \mu u$, $M = \lambda v$ natural charge coordinates
a	assignment (the "current arithmetic value"); on $\mathbb{H}^2$, $a = -x/y$
$\mathcal{A} = \mathbb{R}^2_{(A,M)}$	accumulative commutative space; oriented by $\mathrm{d}A \wedge \mathrm{d}M$
$\mathcal{C} = \mathbb{R}^3_{(u,v,a)}$	contact model; $\alpha = \mathrm{d}a - \mathrm{d}A - a\,\mathrm{d}M$ in charge coordinates
$\pi,\ Q$	charge projection $\pi(u,v,a) = (A,M)$; Reeb quotient $Q(u,v,a) = (-ae^{-\lambda v}, e^{-\lambda v})$
$q,\ p$	ideal boundary point (asymptote) and ordinary point (finite scale)
$\mathsf{A}_p,\ \mathsf{M}_q$	the additive and multiplicative motions, i.e. translation by p and dilation by e^q
$h(p,q)$	the kernel word of Thm. 8.6
$\Pi_{a,b}$	rank-one idempotent of the bivaluation (a,b)
q^+, q^-	image and kernel directions of a bivaluation (Paper IV)
B_∞	$\mathrm{Stab}_{\mathrm{PGL}_2}(\infty)$, the affine Borel sector

continued

symbol	meaning
J	the inverting element $z \mapsto -1/z$
$\Lambda(\square)$	multiplicative cross-ratio holonomy of a four-step bivaluation loop
$\mathcal{T}(e)$	projective torsion transport of an edge
ℓ_0	short-distance reference length (ultraviolet cutoff)
$\mu,\ \lambda$	arithmetic step constants; $\mu\lambda$ is the contact curvature scalar
κ	interaction curvature coefficient, $\kappa = q_1 q_2$ by Hypothesis 1.3

Two conventions matter and are used without further comment. A history is written chronologically from left to right, as in Paper 0: the word $(\mathsf{M}_q, \mathsf{A}_p)$ first scales by e^q and then translates by p. Charges are additive as a function on the history group: the total charge of a word is the sum of its additive and multiplicative increments (Convention 8.3 of Paper 0). Composite maps act on the assignment by composition in the reverse reading, so $(M_k \circ A_h)(a) = e^q(a + p)$ while $(A_h \circ M_k)(a) = e^q a + p$; their difference is the elementary order defect $p(e^q - 1)$ of Appendix D.3 of Paper 0.

1.4 The five dictionary hypotheses

Everything below is proved either from the AEG statements of Subsection 1.2 or from elementary computation, except at five points. Each of those is declared once here, is used in the sections named, and is not silently converted into a theorem.

Hypothesis 1.1 (Charge–motion dictionary)**.** A charged particle at rest is an ideal-point-to-point history whose additive steps carry the signed charge: the electron is generated by A_{-e}, and the positron by A_{+e}, which by the reflection of Paper 0, Subsection 3.4 satisfies $\mathcal{R} \circ \mathsf{A}_{-e} \circ \mathcal{R}^{-1} = \mathsf{A}_{+e}$ with $\mathcal{R}(x, y) = (-x, y)$ scaling $a \mapsto -a$. A virtual photon exchange is one additive step of one history absorbed by another.

Hypothesis 1.2 (Depth–distance dictionary)**.** The multiplicative depth e^{-M} of a history is the Coulomb kernel up to the ultraviolet reference length,

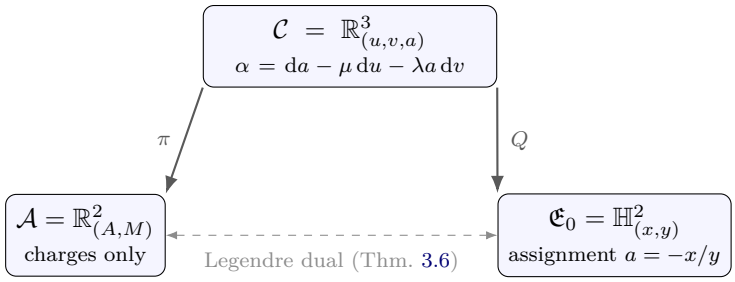
$$e^{-M_*} = \frac{\ell_0}{r} \qquad \Longleftrightarrow \qquad M_* = \log \frac{r}{\ell_0},$$

so that a scale interval $[0, M_*]$ represents a spatial separation r and the ultraviolet end $M = 0$ represents the coincident limit.

Hypothesis 1.3 (Curvature–energy dictionary)**.** The two-body interaction energy is the bilinear pairing of the two charges against the contact curvature. Concretely, writing the additive generator strength $\mu \propto q_1$ and the multiplicative coupling $\lambda \propto q_2$, the curvature scalar of Paper 0, Theorem 9.2 becomes $\omega(D_u, D_v) = \mu\lambda \propto q_1 q_2$, and the potential is the integral of that density over the scale interval selected by Hypothesis 1.2, normalised by the three-dimensional solid angle 4π.

Hypothesis 1.4 (Velocity–tilt dictionary)**.** A charge moving with speed v_{vel} is a history whose depth coordinate advances at rate $\beta = \mathrm{d}M/\mathrm{d}\tau = v_{\mathrm{vel}}/c$, i.e. the direction angle of (26) is the rapidity $\theta = \operatorname{artanh}(v_{\mathrm{vel}}/c)$ of the motion, with c the critical rate at which the tilt saturates.

Hypothesis 1.5 (Holonomy–radiation dictionary)**.** A nonvanishing projective holonomy of a closed history loop, in the sense of the multiplicative component of Paper IV, Theorem 4.11, is the carrier



$\pi(u, v, a) = (A, M)$ forgets the assignment a and keeps the charges;
$Q(u, v, a) = (-ae^{-\lambda v}, e^{-\lambda v})$ forgets the Reeb coordinate u and keeps the state.
The order defect is a tearing along ∂_a and a drift along ∂_u.

Figure 1: The two projections of the contact model, and their Legendre conjugacy. The charge projection π forgets the assignment, so the order defect appears on its fiber ∂_a as a value tearing; the Reeb quotient Q forgets the additive charge, so the same defect appears on its fiber ∂_u as a Reeb drift. Section 3 is the statement that these are two readings of one contact curvature.

of electromagnetic radiation: uniform motion has trivial holonomy and radiates nothing, and an acceleration that moves the kernel coordinate produces a holonomy whose magnitude measures radiated power.

On the status of the hypotheses. None of the five is a theorem of AEG, and none is claimed to be. Each is a dictionary: it names a physical quantity by an AEG object. Hypotheses 1.1 and 1.2 fix what the ends and the scale variable of a history mean; Hypothesis 1.3 fixes what an invariant of two histories means; Hypothesis 1.4 fixes what motion means; Hypothesis 1.5 fixes what a holonomy means. The mathematical content of every section stands without them; the physical conclusions are conditional on them, and are stated in that form. Section 10 collects which conclusion rests on which hypothesis.
One further identification is made later and is not counted among the five because it is a substitution rather than a dictionary: in Section 9 the assignment slot of the contact form is filled by the scale-corrected radial momentum $r\,p_r$ (Lemma 9.2). It is flagged there.

1.5 Roadmap

source note	section	what it contributes
费曼图.md	§2	the $p_0 \to p$ family as an operator fiber; $\mathcal{A}$ fan-out and bare-charge divergence; Reeb-fiber heights
勒让德形式.md	§3	the exponential Legendre identity and the dual tearing theorem
电动力学讨论-01.md	§4	Coulomb's sign rule from the sign of $q_1 q_2$; the dipole loop; the three geometric readings
电动力学讨论-02.md	§5, §8	direction angle and the Ampère coefficient; uniform motion in ripple geometry; Møller versus Bhabha sectors
电动力学讨论-02.md, 电动力学讨论-03.md	§6, §7	moving poles, cross-ratio holonomy, the torsion transport and its Borel reduction

continued

source note	section	what it contributes
电动力学讨论-03.md	§9	elliptic fixed points, Reeb holonomy as action, Bohr–Sommerfeld quantization
—	§10	dictionary table, status ledger, open ends
—	App. A–D	exact computations and the register of verified identities

2 Ideal-Point Feynman Histories

本节把「从理想点 q 到普通点 p 的一族穿线路径」严格写下来，并回答一个具体问题：既然这些路径的起点终点都一样，它们在费曼图、$\mathcal{A}$、切触空间三个视角下分别是什么？结论是：它们恰好是算子求值映射的一条纤维 $\rho^{-1}(g_{p_0\to p})$；在 $\mathcal{A}$ 里摊成一条水平线段（裸荷随 $M\to\infty$ 指数发散），在切触空间里则落在同一根 Reeb 纤维的**不同高度**上。本节只用 Paper 0 的第 7–9 章，不需要任何物理字典。

2.1 Two kinds of ideal point

Let $\mathfrak{E}_0 = (\mathbb{H}^2, g_{\mu,\lambda}, a)$ be the basic model with $a(x,y) = -x/y$. In horocyclic coordinates $u = x$ and $y = e^{\lambda v}$ one has $a = -ue^{-\lambda v}$, and the two affine motions of Paper 0, Proposition 3.6, read

$$\mathsf{X}_s(x,y) = (x - sy,\, y), \qquad \mathsf{Y}_k(x,y) = \Big(x, \frac{y}{k}\Big), \qquad a\circ\mathsf{X}_s = a + s, \quad a\circ\mathsf{Y}_k = ka. \tag{1}$$

The ideal boundary $\partial_\infty\mathbb{H}^2 = \mathbb{R}\cup\{\infty\}$ carries two kinds of point, and the distinction is used constantly below.

(i) The point at infinity, $q = i\infty$, i.e. $y\to+\infty$ or $M = \lambda v\to-\infty$. This is the common base point of the Borel subgroup $B_\infty = \mathrm{Stab}_{\mathrm{PGL}_2}(\infty)$: every affine motion fixes it, so it is the natural origin of an affine history, and in the physical reading it is the ultraviolet, asymptotically free end.

(ii) A finite ideal point $q = x_\infty\in\mathbb{R}$, i.e. $y\to 0^+$ or $M\to+\infty$. This is the infrared end, reached by unbounded positive scaling Y_{e^M} with $M\to+\infty$.

Remark 2.1 (Ultraviolet versus infrared end). Whether one marches from $q = i\infty$ inward to p, or reads the same word backwards from p towards a finite ideal point, is a choice of direction, not a choice of geometry: the two readings are the two ends of one word. Paper 0 keeps both ends in view because the multiplicative charge is the only charge that descends to the motion group (Theorem 8.6), so only M can be compared across histories without an order convention.

2.2 The family of histories is one operator fiber

Remark 2.2 (The ideal point is a direction, not a starting point). B_∞ fixes $q = i\infty$, so no affine motion carries q to an interior point of $\mathbb{H}^2$: a history cannot literally start at the ideal point. What the phrase "a history from q to p" means in this note, and what it means in Paper 0's grid, is a history that starts at an ordinary point p_0 deep in the ultraviolet region — large y_0, near q — and ends at p. The formulation below therefore fixes p_0 explicitly. Its distance from q enters only through the total multiplicative charge, and every invariant used in this note is a function of the charges, not of how deep the start was placed.

Fix an ordinary starting point $p_0 = (x_0, y_0) \in \mathbb{H}^2$ and an ordinary terminal point $p = (x_p, y_p) \in \mathbb{H}^2$. A *comb history* from p_0 to p is a finite chronological word

$$\gamma = (\mathsf{M}_{m_1}, \mathsf{A}_{t_1}, \mathsf{M}_{m_2}, \mathsf{A}_{t_2}, \dots, \mathsf{M}_{m_n}, \mathsf{A}_{t_n}), \qquad m_i, t_i \in \mathbb{R}, \tag{2}$$

read from left to right, whose composite motion carries p_0 to p. Here M_m denotes scaling by e^m and A_t translation by t; the word is the left-expanded comb of Paper 0, Subsection 2.1. In the comb notation of this section we set the multiplicative step constant to $\lambda = 1$, so that the log-scale parameter v and the multiplicative charge $M = \lambda v$ coincide; the general case is recovered by the substitution $M = \lambda v$, and every formula below is homogeneous in the charges.

Proposition 2.3 (The $p_0 \to p$ family is one operator fiber)**.** *Let $g_{p_0 \to p} \in B_\infty$ be the affine motion carrying p_0 to p. Then γ in (2) carries p_0 to p if and only if its composite is $g_{p_0 \to p}$. Consequently the histories joining p_0 to p are exactly one fiber of the evaluation map $\rho : \mathrm{Hist} \to \mathrm{Aff}^+(1, \mathbb{R})$ of Paper 0, Section 7.*

Proof. B_∞ acts simply transitively on $\mathbb{H}^2$: it acts transitively because Y_k and X_s of (1) already carry any point to any other, and freely because a nonidentity affine map has at most one fixed point in $\mathbb{H}^2$ (Paper 0, Proposition 5.4 with $b = 0$; the pure translation has none). Hence exactly one affine motion carries p_0 to p, and a word realizes that motion exactly when it equals $g_{p_0 \to p}$ in the group. □

Writing $g_{p_0 \to p}(z) = e^{M_*} z + \xi_*$, the two invariants of the fiber are the total multiplicative charge $M_* = \lambda \sum_i m_i$ and the total target translation ξ_*. In terms of the assignment,

$$e^{M_*} = \frac{y_p}{y_0}, \qquad a(p) = e^{M_*} a(p_0) + \xi_* = -\frac{x_p}{y_p}. \tag{3}$$

2.3 Comb diagrams and charge accounting

The affine cocycle of Paper 0, Proposition 7.1, expresses the terminal translation of a comb in two equivalent ways,

$$\xi_\gamma = \sum_{i=1}^{n} t_i \, e^{\,M_{i+1:n}} = e^{M_*} \sum_{i=1}^{n} t_i \, e^{-M_i}, \qquad M_{i:n} = \sum_{j=i}^{n} \lambda m_j. \tag{4}$$

Read as a diagram, (4) is a time-ordered Feynman diagram with a single fermion line.

(a) The vertical segments M_{m_i} are free propagation between vertices: they advance the scale by e^{m_i} and cost no additive charge.

(b) The horizontal segments A_{t_i} are vertex insertions at the scale shell M_i: each emits or absorbs one quantum and adds t_i to the running value.

(c) The second form of (4) says that a bare vertex inserted at depth M_i contributes $t_i e^{-M_i}$ to the source-normalized charge and is then transported to the target frame by e^{M_*}.

Proposition 2.4 (The distributive law is the renormalisation resummation)**.** *Fix two vertices at distinct depths. Then for every $M_1 < M_*$ and t_1,*

$$(\mathsf{M}_{M_* - M_1}, \mathsf{A}_{t_1}) \quad \textit{and} \quad (\mathsf{A}_{t_1 e^{M_* - M_1}}, \mathsf{M}_{M_* - M_1})$$

are histories with the same terminal operator but different total charges. The difference of their additive charges is the elementary order defect $t_1 \, (e^{M_* - M_1} - 1)$.

Proof. Both words evaluate to the affine map $z \mapsto e^{M_*} z + e^{M_* - M_1} t_1$, which is the content of the pointwise identity $k(z+p) = kz + kp$ of Paper 0, Subsection 2.3. Their additive charges are t_1 and $t_1 e^{M_* - M_1}$, whose difference is as displayed; the multiplicative charges agree. □

The proposition is the algebraic form of the statement that a large bare vertex inserted deep in the ultraviolet can be transported to the infrared as a small effective vertex. In field-theoretic language the two words are the same renormalized amplitude with the counterterm absorbed, and the charge difference is the order defect that the $\mathcal{A}$ will record as an area.

2.4 The image in the accumulative commutative space

Commutativizing forgets the order of the steps and keeps the two total charges (Paper 0, Definition 8.1 and Proposition 8.5). For the $p_0 \to p$ family this has an immediate consequence.

Proposition 2.5 (The family fans out into a horizontal segment)**.** *Let $\{\gamma\}$ be the histories of Proposition 2.3. Their $\mathcal{A}$ charge chains C_γ all terminate on the horizontal line $M = M_*$, where M_* is fixed by (3), and they satisfy the single scalar constraint*

$$\hat{\xi}(p) := \int_{C_\gamma} e^{-M} \mathrm{d}A = \xi_* \, e^{-M_*} + a(q) \tag{5}$$

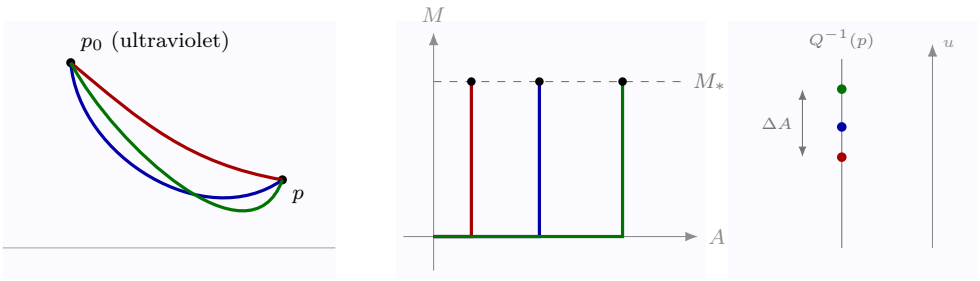
of Paper 0, Proposition 8.2. Along that line the total additive charge A_γ is not determined by p: a history that inserts its vertices early needs $A_\gamma \to$ small, and one that inserts them late needs $A_\gamma \asymp e^{M_*} \hat{\xi}(p)$.

Proof. The multiplicative charge is determined by the endpoint by Theorem 8.6(3). Proposition 8.2 identifies the weighted integral with the source-normalized terminal translation, which is fixed by (3) because both ends are. The two extremes are the two words of Proposition 2.4: inserting at $M = 0$ costs $A = t_1$ and inserting at $M = M_*$ costs $A = t_1 e^{M_*}$. □

Corollary 2.6 (Two readings of the same divergence)**.** *The quantity A_γ — the total additive charge, which the $\mathcal{A}$ keeps as its abscissa — is not a function of the physical endpoint, and it grows like e^{M_*} for histories whose insertions are pushed to the infrared end. The weighted quantity $\hat{\xi}(p)$ is a function of the endpoint and stays finite.*

Warning 2.7 (What this does and does not explain). Corollary 2.6 says that the unweighted sum of insertions over a scale range diverges like e^M while its e^{-M}-weighted version does not. That is exactly the algebraic shape of the statement that a bare quantity regularized at a short distance diverges while a renormalized quantity does not, and the exponential weight is the usual scale factor. It is *not* a proof of renormalizability of any field theory, and no loop integral is computed here. The correspondence is between one exact AEG identity and one structural feature of the perturbative expansion.

When two histories of the family also share their additive charge, they close up in $\mathcal{A}$, and the weighted torsion–Stokes theorem of Paper 0, Theorem 8.9, applies: the relative defect is $\tau(\gamma, \delta) = \iint_{\Sigma_{\gamma\delta}} e^{M_* - M} \mathrm{d}A \wedge \mathrm{d}M$. For two such histories to *also* reach the same p, that torsion must vanish. In the physical reading this is the statement that cancellations between early and late insertions — the finite remainders of renormalization and the Ward-type identities — are exactly the vanishing of a signed weighted area. The elementary case is the kernel word of Theorem 8.6: the two histories $\gamma_1 = (\mathsf{M}_q, \mathsf{A}_p)$ and $\gamma_2 = (\mathsf{A}_{e^{-q}p}, \mathsf{M}_q)$ reach the same state and differ in additive charge by $p(1 - e^{-q})$, which is exactly the weighted area they enclose (Lemma 3.8).



(a) $\mathbb{H}^2$ (b) $\mathcal{A}$ (c) Reeb fiber

Figure 2: Three images of one family of histories. (a) In $\mathbb{H}^2$ the histories have the same two endpoints and therefore the same composite operator. (b) In $\mathcal{A}$ they share the multiplicative charge M_* and differ in additive charge A_γ, subject to the weighted constraint (5). (c) In $\mathcal{C}$ they terminate on one Reeb fiber over p at distinct heights $u_\gamma = A_\gamma/\mu$; the height spread equals the hyperbolic area enclosed by the projected paths (Theorem 3.6).

2.5 The image in the contact model: Reeb heights

Lift each history horizontally to $\mathcal{C}$: a curve $\tilde{\gamma}$ with $\alpha(\dot{\tilde{\gamma}}) = 0$. By Paper 0, Theorem 3.9 and the coordinate form of the horizontal fields, this is the ordinary differential equation $\mathrm{d}a = \mu\,\mathrm{d}u + \lambda a\,\mathrm{d}v$, and by Proposition 9.8(2) its image under Q is the arithmetic motion itself.

Proposition 2.8 (Same state, different Reeb height)**.** *Let $\tilde{\gamma}_1, \tilde{\gamma}_2$ be horizontal lifts of two histories of the $p_0 \to p$ family, starting at the same point of $\mathcal{C}$. Then $Q \circ \tilde{\gamma}_1$ and $Q \circ \tilde{\gamma}_2$ end at the same $p \in \mathbb{H}^2$, while*

$$u(\tilde{\gamma}_1) - u(\tilde{\gamma}_2) = \frac{1}{\mu}(A_{\gamma_1} - A_{\gamma_2}).$$

By Theorem 9.9 this height difference equals $-\iint_{\Omega_{12}} \mathrm{d}a \wedge \mathrm{d}M = -\mu\lambda\,\mathrm{Area}_g(\Omega_{12})$, the hyperbolic area enclosed by the two projected paths in $\mathbb{H}^2$.

Proof. Horizontal lifts of paths with the same endpoints in $\mathbb{H}^2$ end on the same fiber of Q because Q is the projection; the height difference is the statement of Theorem 9.9(2), and the additive charge is $A = \mu u$ by the natural charge coordinates of Appendix D.4. □

Remark 2.9 (Action and phase). The Reeb coordinate u is the coordinate along which the contact form evaluates to 1: $\alpha(R) = 1$ with $R = -\frac{1}{\mu}\partial_u$ (Appendix D.4 of Paper 0). In the geometric quantization of a contact manifold, the Reeb direction is the circle fiber of the prequantum bundle and its coordinate is the phase of the wave function [10, 11]. Proposition 2.8 therefore says that histories reaching the same state with different Reeb heights differ by a phase. Summing over the fiber with weights $e^{\mathrm{i}A_\gamma}$ is then the natural candidate for a path integral in this picture. Nothing in this note depends on that candidate, which is recorded as a target rather than as a result.

2.6 Radix shells

If the scaling steps are quantized on a k-ary grid and each shell either emits or does not, the word becomes a Horner expansion and the terminal translation is

$$\xi = \sum_{i=1}^{n} d_i\, k^{\,n-i}, \qquad d_i \in \{0, 1, \ldots, k-1\}, \tag{6}$$

with the counting statements of Paper IV, Proposition 10.1 and Theorem 10.3. In the physical reading $d_i \neq 0$ at shell i records a virtual process at the Wilsonian momentum shell k^{n-i}, and the terminal value is built up shell by shell from the ultraviolet end. Once signed and continuous increments are allowed, the number of histories sharing one terminal operator is infinite, which is the content of Proposition 2.3.

2.7 Panorama

carrier	AEG object	Feynman reading	invariant that survives
$\mathfrak{E}_0$	broken-line family from $q = \infty$ to p; one fiber $\rho^{-1}(g_{p_0 \to p})$	time- and scale-ordered comb diagrams from the asymptotic state to the dressed state	endpoint assignment $a(p)$ and total scale e^{M_*} fixed; the order of insertions is retained
$\mathcal{A}$	all chains end on $M = M_*$; abscissa $A_\gamma = \sum t_i$ free subject to (5)	bare versus renormalized charge; ultraviolet sensitivity of the unweighted sum	$\hat{\xi}(p) = \int_C e^{-M} \mathrm{d}A$; and $\tau(\gamma, \delta) = 0$ when both charge and endpoint agree
$\mathcal{C}$	Legendrian lift family; endpoints on distinct heights of $Q^{-1}(p)$	different histories for one physical state, carrying different phases	Reeb drift $A_{\gamma_1} - A_{\gamma_2} = -\mu\lambda \operatorname{Area}_g(\Omega_{12})$

Section 3 isolates the last row as a theorem about pairs of histories and then re-reads the first row as its conjugate.

3 The Legendre Form: Charge Projection versus Reeb Quotient

本节回答一个自然的问题：切触空间里那两个投影 π（忘掉赋值 a）与 Q（忘掉 Reeb 荷 u）之间的「勒让德对偶」，究竟与力学里联系 Lagrange 量和 Hamilton 量的**勒让德变换**是不是同一件事？答案是：在标准切触流形上，它就是同一个操作的两个 Legendre 叶层。由此得到本文的第一个真正的新命题：**双重撕裂定理**——固定荷看「赋值撕裂 Δa」(Paper 0 定理 8.9)，固定状态看「Reeb 纤维漂移 ΔA」(Paper 0 定理 9.9)，两者是同一个切触曲率的两个共轭读法。本节末尾用核字 $h(p, q)$ 做两条路径的现场验算，结果一字不差是 $p(1 - e^{-q})$。

3.1 Why the mechanical Legendre transform is a contact statement

In analytical mechanics with one degree of freedom, let q be the coordinate, p the momentum, and S the action, so that (q, p, S) are coordinates on the first jet bundle $J^1(\mathbb{R}) \cong \mathbb{R}^3$. The canonical contact form is

$$\alpha_{\mathrm{mech}} = \mathrm{d}S - p\,\mathrm{d}q = 0. \tag{7}$$

Using $\mathrm{d}(pq) = p\,\mathrm{d}q + q\,\mathrm{d}p$ and putting $\tilde{S} := pq - S$, the same form reads

$$\alpha_{\mathrm{mech}} = -\mathrm{d}\tilde{S} + q\,\mathrm{d}p = 0. \tag{8}$$

Equations (7) and (8) are the same one-form written through two different choices of potential, and the two choices are related by the Legendre transformation. Geometrically the content is that one contact three-manifold carries two Legendrian fibrations: the "coordinate" projection $(q, p, S) \mapsto (q, S)$ in which $p\,\mathrm{d}q$ is read as the accumulated quantity, and the "momentum" projection in which $q\,\mathrm{d}p$ is. Nothing about the contact structure changes; what changes is which variable is declared to be a function of the others.

Remark 3.1 (What the Legendre transformation does not do). The passage from (7) to (8) uses only the product rule. It is therefore available wherever the product exists; the analytic content of the classical Legendre transformation (convexity, invertibility of $p = \partial S/\partial q$) is not used and is not needed here.

3.2 The arithmetic contact form and its two potentials

In the natural charge coordinates $A = \mu u$ and $M = \lambda v$ of Paper 0, Appendix D.4, the arithmetic contact form is

$$\alpha = \mathrm{d}a - \mathrm{d}A - a\,\mathrm{d}M, \tag{9}$$

and the global affine change of coordinates

$$x_{\mathrm{D}} = M, \qquad y_{\mathrm{D}} = a, \qquad z_{\mathrm{D}} = a - A \tag{10}$$

puts it in standard Darboux form $\alpha = \mathrm{d}z_{\mathrm{D}} - y_{\mathrm{D}}\,\mathrm{d}x_{\mathrm{D}}$ (Appendix D.4). Comparing (10) with (7) identifies the correspondence

$$S \;\longleftrightarrow\; a - A, \qquad p \;\longleftrightarrow\; a, \qquad q \;\longleftrightarrow\; M. \tag{11}$$

The current assignment a plays the role of the momentum, and the logarithmic scale M plays the role of the coordinate.

Lemma 3.2 (Legendre-dual form of α)**.** *With $\tilde{S} := aM - a + A$ one has the identity*

$$\alpha = -\mathrm{d}\tilde{S} + M\,\mathrm{d}a \qquad \text{on } \mathcal{C}. \tag{12}$$

Proof. Expanding $-\mathrm{d}(aM - a + A) + M\,\mathrm{d}a$ gives $-(a\,\mathrm{d}M + M\,\mathrm{d}a - \mathrm{d}a + \mathrm{d}A) + M\,\mathrm{d}a = \mathrm{d}a - \mathrm{d}A - a\,\mathrm{d}M$. □

Remark 3.3 (A correction to the working note). The working note from which this section is drawn displayed the same step with the coefficient $1 + M$ on $\mathrm{d}a$, that is $\alpha = -\mathrm{d}(A + aM - a) + (1 + M)\mathrm{d}a$. Expanding shows that this has an extra $\mathrm{d}a$ and is not α. The correct dual potential is $A + aM - a$ with momentum coefficient M, as in (12). The error does not propagate: every later identity in the note uses the exponential form (13) below, which is correct as written. Check V5.1–V5.2 of Appendix D records both.

The linear Legendre transformation (12) is not the most intrinsic one available here, because the multiplicative motion of AEG is the exponential e^M rather than the addition of M. Rescaling the contact form by the nowhere-vanishing factor e^{-M} — which leaves the horizontal distribution $\ker\alpha$ untouched — gives a second, multiplicative dual.

Lemma 3.4 (Exponential Legendre identity)**.** *On $\mathcal{C}$,*

$$e^{-M}\alpha = \mathrm{d}(e^{-M}a) - e^{-M}\mathrm{d}A. \tag{13}$$

Proof. $e^{-M}\alpha = e^{-M}(\mathrm{d}a - a\,\mathrm{d}M) - e^{-M}\mathrm{d}A$ and $\mathrm{d}(e^{-M}a) = e^{-M}\mathrm{d}a - ae^{-M}\mathrm{d}M$. □

Remark 3.5 (Three names for one quantity). The quantity $\hat{a} := e^{-M}a$ occurring in (13) has three readings. It is the source-normalized assignment of Paper 0, Proposition 8.2; it is minus the horizontal coordinate of the basic hyperbolic model, because under the Reeb quotient $Q(u, v, a) = (-ae^{-\lambda v}, e^{-\lambda v})$ one has $\hat{a} = -x$; and it is, by Lemma 3.4, the potential whose differential measures the additive charge in the target frame. Paper 0, Theorem 9.9 is exactly the statement that these three readings agree.

3.3 The dual tearing theorem

The two readings of Lemmas 3.2 and 3.4 become two comparison experiments. The statement below is Paper 0, Theorem 9.9; it is reproduced here in the notation of this note because every later section uses it.

Theorem 3.6 (Dual tearing and Reeb drift)**.** *Let $\tilde{\gamma}_1, \tilde{\gamma}_2$ be piecewise smooth horizontal lifts in $\mathcal{C}$ with a common initial state.*

(1) (Fixed charges: value tearing.) *Suppose the projected charge chains $C_i = \pi \circ \tilde{\gamma}_i$ share their terminal point (A_*, M_*). Then the terminal assignments differ by*

$$a(\tilde{\gamma}_1(1)) - a(\tilde{\gamma}_2(1)) = e^{M_*} \oint_{C_1 - C_2} e^{-M} \mathrm{d}A = \iint_{\Sigma_{12}} e^{M_* - M} \mathrm{d}A \wedge \mathrm{d}M = \tau(\gamma_1, \gamma_2), \tag{14}$$

for any oriented two-chain Σ_{12} in $\mathcal{A}$ with $\partial\Sigma_{12} = C_1 - C_2$. This is the weighted torsion–Stokes formula of Paper 0, Theorem 8.9.

(2) (Fixed state: Reeb drift.) *Suppose instead that the projected paths $\tilde{\gamma}_i = Q \circ \tilde{\gamma}_i$ terminate at the same point $p \in \mathbb{H}^2$, equivalently at the same pair (M_*, a_*). Then the terminals lie on the same Reeb fiber and*

$$\Delta A := A(\tilde{\gamma}_1(1)) - A(\tilde{\gamma}_2(1)) = -\oint_{\tilde{\gamma}_1 - \tilde{\gamma}_2} a \, \mathrm{d}M = -\iint_{\Omega_{12}} \mathrm{d}a \wedge \mathrm{d}M = -\mu\lambda \, \mathrm{Area}_{g_{\mu,\lambda}}(\Omega_{12}), \tag{15}$$

for any oriented two-chain $\Omega_{12} \subset \mathbb{H}^2$ with $\partial\Omega_{12} = \tilde{\gamma}_1 - \tilde{\gamma}_2$.

Proof. (1) Integrate (13) along each horizontal lift; the initial terms cancel on subtraction, and e^{M_*} multiplying the result gives the first equality. The second is Stokes' theorem in $\mathcal{A}$ with the target weight $e^{M_* - M}$, as in Paper 0, Theorem 8.9.

(2) Integrate (9) along each lift to get $A(\tilde{\gamma}_i(1)) - A_0 = (a(\tilde{\gamma}_i(1)) - a_0) - \int_{\tilde{\gamma}_i} a \, \mathrm{d}M$. Under common terminal state the boundary terms cancel on subtraction, giving the line-integral form. Stokes' theorem in $\mathbb{H}^2$ and $d\alpha = -\mu\lambda \, Q^* \mathrm{dvol}_{g_{\mu,\lambda}}$ of Paper 0, Proposition 9.8(5) give the area form. □

Remark 3.7 (Why the two projections must be conjugate). π forgets the assignment fiber ∂_a and Q forgets the Reeb fiber ∂_u. These are different directions of one three-manifold, so a comparison that closes in one projection necessarily fails to close in the other: a charge-compatible pair stays open on $\mathbb{H}^2$, and a state-compatible pair closes on $\mathbb{H}^2$ and opens along the Reeb fiber. That complementarity — not an analogy — is the content of the word "duality" here.

3.4 A worked pair: the kernel word

The smallest nontrivial instance is the element of $\ker \nu$ exhibited in Paper 0, Theorem 8.6,

$$h(p, q) = \mathsf{M}_q \, \mathsf{A}_p \, \mathsf{M}_{-q} \, \mathsf{A}_{-e^{-q}p}, \tag{16}$$

whose two halves are the prefixes $\gamma_1 = (\mathsf{M}_q, \mathsf{A}_p)$ and $\gamma_2 = (\mathsf{A}_{e^{-q}p}, \mathsf{M}_q)$. Both carry $(M, a) = (0, 0)$ to the same terminal state with $M_* = q$ and $a_* = p$.

Lemma 3.8 (Reeb drift of the kernel word)**.** *For γ_1, γ_2 as above,*

$$\Delta A = A(\gamma_1) - A(\gamma_2) = p - e^{-q}p = p(1 - e^{-q}),$$

and the two sides of (15) both evaluate to $p(1 - e^{-q})$. The weighted torsion of (14) for the same pair vanishes, as it must, since the terminal assignments agree.

Proof. The bare additive charges are $A(\gamma_1) = p$ and $A(\gamma_2) = e^{-q}p$. On the state side, γ_1 is the lower broken line $(0,0) \to (q,0) \to (q,p)$ with $a \equiv 0$ along its horizontal leg, so $\int_{\gamma_1} a\,\mathrm{d}M = 0$, while γ_2 follows $a(M) = pe^{M-q}$ and $\int_{\gamma_2} a\,\mathrm{d}M = \int_0^q pe^{M-q}\mathrm{d}M = p(1-e^{-q})$. Hence $-\big(\int_{\gamma_1} - \int_{\gamma_2}\big)a\,\mathrm{d}M = p(1-e^{-q})$, in agreement with the charge difference. The weighted area route gives $\int_0^q\int_0^p e^{-M}\mathrm{d}A\,\mathrm{d}M = p(1-e^{-q})$ as well. The verification is Check V6.1–V6.3. □

This is the identity that Paper 0, Appendix D.2 records as the archetype of "zero total charge is weaker than operator neutrality": the loop $\ell = (\mathsf{A}_p, \mathsf{M}_q, \mathsf{A}_{-p}, \mathsf{M}_{-q})$ has total charge $(0,0)$ yet $\tau(\ell, \varnothing) = p(1-e^{-q})$. In Section 4 this lemma is read as the binding energy of a charge dipole, and in Section 6 it is read as the phase carried by a virtual pair.

3.5 The duality table

	reading A: charge projection π	**reading B: Reeb quotient Q**
map	$\mathcal{C} \to \mathcal{A}_{(A,M)}$, forgets a	$\mathcal{C} \to \mathfrak{E}_0$, forgets $u = A/\mu$
form used	$e^{-M}\alpha = \mathrm{d}(e^{-M}a) - e^{-M}\mathrm{d}A$	$\alpha = -\mathrm{d}A + (\mathrm{d}a - a\,\mathrm{d}M)$
closure hypothesis	same terminal charges (A_*, M_*)	same terminal state $p = (M_*, a_*)$
vertical defect	value tearing Δa	Reeb drift $\Delta A = \mu\Delta u$
value	$\Delta a = \iint_\Sigma e^{M_*-M}\mathrm{d}A \wedge \mathrm{d}M$	$\Delta A = -\iint_\Omega \mathrm{d}a \wedge \mathrm{d}M = -\mu\lambda\,\mathrm{Area}_g(\Omega)$
AEG source	Theorem 8.9, Prop. 8.2	Theorem 8.6, Prop. 9.8, Theorem 9.9

3.6 The open thread: from the affine sector to the projective group

Both readings above live in the affine sector: the horizontal lifts are lifts inside B_∞, and the Riccati structure of the assignment along a flow is hidden. If one allows the second-slot division c/z, i.e. the inverting element $J(z) = -1/z$, then the same one-form acquires a quadratic term and the assignment obeys a Riccati rather than a linear equation. Section 6 carries out exactly that step, and shows that the additional datum which then appears — the kernel coordinate b of the bivaluation — is what makes a closed history carry a non-telescoping holonomy.

4 Electrostatics: Coulomb's Law from Contact Curvature

本节是全文的第一次「校准」：把第 3 节的双重撕裂定理用到两个静止电荷上，看能不能不借助任何外加的闵氏度规，直接从 AEG 的几何量里读出库仑定律的 **符号规律**（同号相斥、异号相吸）。做法是：把两个电荷写成两次加法步 q_1, q_2，相互作用能取为切触曲率的双线性配对 $\omega(D_u, D_v) = \mu\lambda$，于是 $\kappa = q_1 q_2$ 的符号就是势能的符号；而正电子的 $+e$ 由 Paper 0 第 3.4 小节的反射算子给出，它恰好翻转了 2-链的定向（附录 D.1）。本节 4.6 小节逐条列出哪些是已证事实、哪些是字典假设。

4.1 Stage and dictionary

Work on $\mathfrak{E}_0$ and its contact model with the natural charge coordinates (A, M) of (9). Two charges at rest are two ideal-point histories whose multiplicative charges are equal, because both sit at the same spatial separation from the observation point in the Hypothesis 1.2 sense,

$$e^{-M_*} = \frac{\ell_0}{r} \qquad \Longleftrightarrow \qquad M_* = \log\frac{r}{\ell_0}, \tag{17}$$

so that $M = 0$ is the ultraviolet reference (coincident limit) and $M_* \to \infty$ is infinite separation. Charge 1 injects its additive step at the ultraviolet end; the field it creates at the location of charge 2 is the assignment reached after the scale step M_*,

$$a_1(p) = q_1\, e^{-M_*} = \frac{q_1 \ell_0}{r}, \tag{18}$$

which under (17) is the Coulomb kernel up to the reference length.

The two signs of charge are the two signs of the additive generator.

Lemma 4.1 (Reflection conjugates the electron step into the positron step)**.** *Let $\mathcal{R}(x, y) = (-x, y)$ be the reflection of Paper 0, Subsection 3.4, which acts on the assignment by $a \mapsto -a$, and let X_s be the translation* (1)*. Then*

$$\mathcal{R} \circ \mathsf{X}_{-e} \circ \mathcal{R}^{-1} = \mathsf{X}_{+e}, \qquad \textit{equivalently} \qquad a \circ \mathcal{R} \circ \mathsf{X}_{-e} \circ \mathcal{R}^{-1} = a + e.$$

Proof. $\mathcal{R}^{-1} = \mathcal{R}$, and for a point (x, y), $\mathcal{R}(x, y) = (-x, y)$, then $\mathsf{X}_{-e}(-x, y) = (-x + ey, y)$, then $\mathcal{R}(-x + ey, y) = (x - ey, y) = \mathsf{X}_{+e}(x, y)$. □

Warning 4.2 (The reflection is not one of the four operations)*.* Paper 0, Subsection 3.4 is explicit that the reflection is not itself one of the four arithmetic operations: it appears because the assignment of the model is signed, and it is the elementary instance of the second-slot subtraction $c - z$. Multiplication by a negative constant is the reflection composed with a positive scaling. Lemma 4.1 must be read with that restriction: what is conjugated is the additive motion, and the conjugation uses a map that is outside the positive affine group.

4.2 The exchange and the interaction energy

Consider the two histories of the exchange: charge 1 emits at the ultraviolet end and the field is transported to charge 2, and the conjugate history in which charge 2 emits and the field is transported back. They are the histories γ_1 and γ_2 of Lemma 3.8, with additive steps q_1, q_2 in place of p and multiplicative charge M_* in place of q.

The interaction energy is then obtained in either of two equivalent ways.

(a) *Pairing of field and copoint.* The displacement of the assignment produced by charge 1 at the location of charge 2 is $\Delta a_1(r) = q_1 \ell_0 / r$; contracting it with the dual functional carried by charge 2 gives, after the 4π solid-angle normalisation,

$$V(r) = \frac{1}{4\pi \ell_0}\, q_2\, \Delta a_1(r) = \frac{q_1 q_2}{4\pi r}. \tag{19}$$

(b) *Contact curvature density.* Do not pair externally. Instead let the additive generator strength be $\mu \propto q_1$ and the multiplicative coupling be $\lambda \propto q_2$. Then the curvature scalar of Paper 0, Theorem 9.2 is

$$\omega(D_u, D_v) = \mu\lambda \;\propto\; q_1 q_2 =: \kappa, \tag{20}$$

and integrating the density over the scale interval $[0, M_*]$ selected by (17) gives the same $V(r) = \kappa\, e^{-M_*}/(4\pi\ell_0) = \kappa/(4\pi r)$.

Reading (b) is the one that carries the sign for free: by Theorem 9.2 the curvature scalar of the horizontal distribution is literally $\mu\lambda$, and by Remark 9.5 of Paper 0 it is an intrinsic invariant of the distribution even though its numerical value depends on the choice of horizontal metric. The sign of κ is therefore the sign of the product of the two charge steps, and that is the whole of the sign rule.

Where Hypothesis 1.3 enters. Theorem 9.2 supplies $\omega(D_u, D_v) = \mu\lambda$ as an exact statement about the contact distribution. It does not by itself say that λ is proportional to a second charge, nor that the integral of the density over a scale interval is an energy. Both are the content of Hypothesis 1.3, and the force law below is conditional on it. What is *not* hypothetical is the sign structure: once $\kappa = q_1 q_2$, the two cases differ by the sign of a product of two real numbers, and no further choice is available.

4.3 Like charges: the two-electron case

Take $q_1 = q_2 = -e$. Then

$$\kappa_{ee} = q_1 q_2 = (-e)(-e) = +e^2 > 0, \tag{21}$$

which is the same sign as for two positive charges. Geometrically this is Lemma 4.1 applied twice: the two reflections of the second slot compose to the identity, $\mathcal{R}^2 = \mathrm{id}$, so the 2-chain keeps its orientation. Paper 0, Appendix D.1 makes the same statement at the level of the oriented rectangle: the pulled-back orientation form is $pq\,\mathrm{d}s \wedge \mathrm{d}t$, so a double sign change leaves it $+pq\,\mathrm{d}s \wedge \mathrm{d}t$.

Integrating the curvature density over the scale interval as in (19),

$$V_{ee}(r) = \frac{\kappa_{ee}}{4\pi\ell_0} e^{-M_*} = +\frac{e^2}{4\pi r} > 0. \tag{22}$$

The generalized force conjugate to the separation r is

$$F_{ee}(r) = -\frac{\mathrm{d}V_{ee}}{\mathrm{d}r} = -\frac{\mathrm{d}}{\mathrm{d}r}\Big(+\frac{e^2}{4\pi r}\Big) = +\frac{e^2}{4\pi r^2} > 0, \tag{23}$$

i.e. directed along increasing r: the charges are pushed apart.

Remark 4.3 (Why a positive potential means repulsion here). The sign convention is the one fixed in Section 1.3: V is the energy of the configuration as a function of the separation, so $F = -\mathrm{d}V/\mathrm{d}r$ is the force conjugate to r. In the $\mathfrak{E}_0$ reading of Subsection 4.5 below, $V_{ee} > 0$ is exactly the statement that the configuration relaxes towards $r \to \infty$, i.e. towards $e^{-M} \to 0$ on the ideal boundary.

4.4 Opposite charges: the electron–positron case

Take $q_1 = -e$ and $q_2 = +e$. By Lemma 4.1 the second charge is the first one conjugated by the reflection, so exactly one side of the 2-chain carries a sign change.

Lemma 4.4 (Orientation reversal for exactly one negative side)**.** *For the oriented singular rectangle $R_{p,q}(s,t) = (sp, tq)$ with $0 \le s, t \le 1$ of Paper 0, Appendix D.1, the pulled-back orientation form is $pq\,\mathrm{d}s \wedge \mathrm{d}t$. Hence the orientation of the rectangle reverses exactly when one of p, q changes sign, and the weighted integral $\int_{R_{p,q}} e^{q-M}\mathrm{d}A \wedge \mathrm{d}M = p(e^q - 1)$ is unchanged.*

Proof. $R_{p,q}^*(\mathrm{d}A \wedge \mathrm{d}M) = pq\,\mathrm{d}s \wedge \mathrm{d}t$; the weight is positive, so the sign of the integral is the sign of pq. The value $p(e^q - 1)$ follows by direct integration as in Appendix D.1. □

Consequently the curvature pairing changes sign,

$$\kappa_{e^+e^-} = q_1 q_2 = (-e)(+e) = -e^2 < 0, \tag{24}$$

and with it the potential and the force:

$$V_{e^+e^-}(r) = -\frac{e^2}{4\pi r} < 0, \qquad F_{e^+e^-}(r) = -\frac{e^2}{4\pi r^2} < 0. \tag{25}$$

The force is along decreasing r: the charges are pulled together.

Two further AEG identities describe the same case.

Proposition 4.5 (The neutral pair is the closed charge loop)**.** *For equal and opposite additive steps the total additive charge vanishes,* $A_{\mathrm{tot}} = (-e) + (+e) = 0$*, and the pair is realized by the closed chronological loop* $\ell = (\mathsf{A}_{+e}, \mathsf{M}_{M_*}, \mathsf{A}_{-e}, \mathsf{M}_{-M_*})$ *of Paper 0, Appendix D.2. Its weights satisfy*

$$\tau(\ell, \varnothing) = e(1 - e^{-M_*}) = e\Big(1 - \frac{\ell_0}{r}\Big),$$

which vanishes as $r \to \ell_0$ *and grows towards* e *as* $r \to \infty$*.*

Proof. The charge computation is Theorem 8.6(2) with $(p, q) \mapsto (e, M_*)$; the evaluation defect is the lemma of Section 3.5. The limits are immediate from (17). □

Remark 4.6 (What the closed loop adds)*.* The loop of Proposition 4.5 is the identity that Paper 0 records to separate "zero total charge" from "operator neutrality". Read through Hypothesis 1.3 it acquires a second reading: the dipole has zero asymptotic charge, but because the $+e$ and $-e$ steps occur at different depths it encloses a signed rectangle in $\mathcal{A}$, and the weighted area of that rectangle is the defect $\tau = e(1 - e^{-M_*})$. Since $e(1 - \ell_0/r)$ increases with r, the configuration lowers its defect by decreasing r: this is the geometric source of the attraction, stated entirely inside AEG up to the identification of τ with an energy.

4.5 Three geometric readings of the sign rule

(1) In $\mathfrak{E}_0$: opening versus closing. Two electrons push the assignment in the same direction, X_{-e} each. To keep the terminal state fixed they need compensating rescaling, and the compensation grows with the scale span M_*; the pair behaves like an opening spring whose relaxation is towards larger r. Electron and positron push in opposite directions, and at $M_* = 0$ the two motions cancel identically, $\mathsf{X}_{+e} \circ \mathsf{M}_0 \circ \mathsf{X}_{-e} = \mathrm{id}$, so the configuration is already neutral in the coincident limit; any nonzero separation inserts a scale step M_* between the two steps and produces the defect $e(1 - e^{-M_*})$ that the pair then reduces by approaching.

(2) In $\mathcal{A}$: divergent stacking versus a closed rectangle. Two like charges give $A_{\mathrm{tot}} = -2e \neq 0$: no closed loop, hence no cancellation, and the additive charge accumulates without bound as $M_* \to \infty$ (Corollary 2.6). Opposite charges give $A_{\mathrm{tot}} = 0$ and the closed rectangle of Proposition 4.5, whose weighted area is the binding defect.

(3) In $\mathcal{C}$: the sign of the curvature and the handedness of the contact volume. The curvature two-form is $\omega = \mu\lambda\, \mathrm{d}u \wedge \mathrm{d}v$ with $\mu\lambda \propto q_1 q_2$; the contact volume form is $\alpha \wedge \mathrm{d}\alpha = \mu\lambda\, \mathrm{d}u \wedge \mathrm{d}a \wedge \mathrm{d}v$ (Paper 0, Propositions 9.1 and 9.6). Changing the sign of exactly one charge therefore reverses the orientation of the contact volume while leaving the horizontal distribution unchanged in rank and integrability. In the reading of Theorem 3.6(2) this is a reversal of the Reeb drift: for like charges the Reeb height of the pair grows with decreasing r, for unlike charges it decreases.

4.6 Status of this section

statement	status
$\mathcal{R} \circ \mathsf{X}_{-e} \circ \mathcal{R}^{-1} = \mathsf{X}_{+e}$	proved (Lemma 4.1); Paper 0, Subsec. 3.4

continued

statement	status
orientation of the rectangle reverses when exactly one side is negative	proved (Lemma 4.4); Paper 0, App. D.1
$(-p)(-q) = +pq$ versus $(-p)(+q) = -pq$ for the curvature pairing	proved once $\kappa = \mu\lambda \propto q_1 q_2$; Paper 0, Props. 9.1, 9.6
the neutral pair is a closed loop with defect $e(1 - e^{-M_*})$	proved; Paper 0, Thm. 8.6, App. D.2
the additive step A_{-e} is a physical charge $-e$, and $\mathcal{R}$ is charge conjugation	Hypothesis 1.1
$e^{-M_*} = \ell_0/r$ is the Coulomb kernel	Hypothesis 1.2
$V = \kappa e^{-M_*}/(4\pi\ell_0)$ is an interaction energy, i.e. $\mu\lambda$ pairing is energy	Hypothesis 1.3
$\tau = e(1 - e^{-M_*})$ is a binding energy	Hypothesis 1.3, via Remark 4.6

The three-dimensional normalisation by 4π is not derived here. It is the solid angle of the unit sphere, inserted because the AEG model is one dimensional in the multiplicative direction while the physical field lives in three space dimensions; the same insertion is used in every section that quotes a force. A genuinely three-dimensional carrier is beyond the scope of the present note and is recorded as an open end in Section 10.

5 Moving Charges: Direction Angle and the Ampère Force

静止电荷只有标量势，运动电荷才出现矢势与磁场。本节把「运动」写成 AEG 里的**方向角** θ：静止时 $\theta = 0$（纯加法步，对应只有时间分量的流），运动时标度与平移同时演化（对应倾斜的世界线）。于是上一节那条「切触联络 1-形式在轨迹上的配对」$\alpha_1(\dot\gamma_2)$ 立刻给出 $\frac{q_1 q_2}{4\pi r}\Big(1 - \frac{\mathbf{v}_1\cdot\mathbf{v}_2}{c^2}\Big)$：第二项就是安培力，并且解释了为什么平行电流相吸、为什么 $v \to c$ 时磁吸引精确抵消静电斥力。本节的符号来源是 Paper 0 定理 3.9；把 θ 认作快度、把 v 认作 $c\tanh\eta$，是本节的第五条字典假设。

5.1 Direction angle and inclined histories

Paper 0, Theorem 3.9, together with equations (21) and (22) of that paper — the angular generator $\Omega_\theta = \mu\cos\theta\, E + \lambda\sin\theta\, H$ and the flow $\mathrm{d}z/\mathrm{d}s = \mu\cos\theta + \lambda z\sin\theta$ — states that for a horizontal curve whose tangent is $\cos\theta\, D_u + \sin\theta\, D_v$ one has

$$\frac{\mathrm{d}u}{\mathrm{d}s} = \cos\theta, \qquad \frac{\mathrm{d}v}{\mathrm{d}s} = \sin\theta, \qquad \frac{\mathrm{d}a}{\mathrm{d}s} = \mu\cos\theta + \lambda a\sin\theta. \tag{26}$$

Two readings are distinguished by θ.

(i) *At rest*, $\theta = 0$: only the additive direction is active, $\mathrm{d}v = 0$. The history is a purely horizontal segment in $\mathcal{A}$, the assignment evolves by translation, and the current has only the "time" component. This is the configuration of Section 4.

(ii) *In motion*, $\theta \neq 0$: the same history simultaneously advances its scale and its value, $\mathrm{d}v \neq 0$, and in $\mathcal{A}$ it is an inclined segment whose slope $\mathrm{d}M/\mathrm{d}\tau$ measures the motion. The stationary points of the assignment therefore also drift: a moving charge changes the scale at which its field is read.

5.2 The indefinite signature of the frame algebra

The velocity dictionary is Hypothesis 1.4. The coefficient $1 - \mathbf{v}_1 \cdot \mathbf{v}_2/c^2$ that appears below is not put in by hand: it is what a one-form pairing produces when the vector it is paired with has an indefinite (Lorentzian) inner product. Paper 0, Remark 9.7, is careful to say that the curvature two-form ω is intrinsic while its scalar value depends on the chosen horizontal metric; the choice between a Euclidean and a Lorentzian horizontal metric is therefore exactly the choice between electrostatics and electrodynamics.

Lemma 5.1 (The frame algebra is indefinite)**.** *In the Lie algebra* $\mathfrak{sl}_2(\mathbb{R})$ *with* $X = \begin{pmatrix} m & a_+ \\ a_- & -m \end{pmatrix}$ *one has* $\det X = -(m^2 + a_+ a_-)$*. Consequently the quadratic form* $(m, a_+, a_-) \mapsto \det X$ *has signature* $(2, 1)$*: the direction of the scale rate* m *enters with the opposite sign to the mixed translation entries.*

Proof. $\det X = -m^2 - a_+ a_-$ is a direct computation. The form $-m^2 - a_+ a_-$ is negative definite on the line $a_\pm = 0$ and of split sign on the plane $m = 0$, whence signature $(1, 2)$ up to an overall sign, i.e. $(2, 1)$ after overall negation. □

Remark 5.2. Lemma 5.1 is the algebraic shadow of the statement that the time direction and the space directions of a moving source cannot be treated on the same footing. It is recorded here because it explains why the coefficient in (29) is a difference rather than a sum. No identification of m with a physical time direction is made; that identification is Hypothesis 1.4.

5.3 The contact pairing for two moving charges

Let charge 1, with charge q_1 and velocity $\mathbf{v}_1$, excite the gauge potential read as the contact one-form

$$\alpha_1 = \phi_1 \, \mathrm{d}t - \mathbf{A}_1 \cdot \mathrm{d}\mathbf{x} = \frac{q_1}{4\pi r}\Big(\mathrm{d}t - \mathbf{v}_1 \cdot \mathrm{d}\mathbf{x}\Big). \tag{27}$$

In AEG terms, $\mathrm{d}t$ is the pure additive frame direction and $\mathbf{v}_1 \cdot \mathrm{d}\mathbf{x}$ is the synchronous drift of the scale coordinate caused by the motion, i.e. the β_1 multiple of the multiplicative direction of Hypothesis 1.4. Let charge 2 follow its own inclined history with tangent $\dot\gamma_2 = (1, \mathbf{v}_2)$.

Proposition 5.3 (Ampère coefficient from a one-form pairing)**.** *Under Hypotheses 1.1–1.4, the interaction Lagrangian density of the two charges is the evaluation of the contact form on the second history's tangent,*

$$L_{\mathrm{int}} = q_2 \, \alpha_1(\dot\gamma_2) = \frac{q_1 q_2}{4\pi r}\Big(1 - \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{c^2}\Big), \tag{28}$$

so that the effective potential and force are

$$V_{\mathrm{eff}}(r) = \frac{q_1 q_2}{4\pi r}\Big(1 - \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{c^2}\Big), \qquad F(r) = -\frac{\partial V_{\mathrm{eff}}}{\partial r} = \frac{q_1 q_2}{4\pi r^2}\Big(1 - \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{c^2}\Big). \tag{29}$$

Proof. Insert $\dot\gamma_2 = (1, \mathbf{v}_2)$ into (27) to get $\alpha_1(\dot\gamma_2) = \frac{q_1}{4\pi r}(1 - \mathbf{v}_1 \cdot \mathbf{v}_2)$ (with c restored in the second term), and multiply by the second charge. This is the interaction Lagrangian density; for a static pair it coincides with the potential energy of (19) at $\mathbf{v}_1 = \mathbf{v}_2 = 0$, so no sign change is needed in passing to V_{eff}. Differentiating in r gives (29). □

The coefficient splits into the static Coulomb part of Section 4 and a purely motional part:

$$F(r) = \underbrace{\frac{q_1 q_2}{4\pi r^2}}_{\text{static}} - \underbrace{\frac{q_1 q_2 \, (\mathbf{v}_1 \cdot \mathbf{v}_2)}{4\pi c^2 r^2}}_{\text{motional}}. \tag{30}$$

Corollary 5.4 (Parallel currents attract the electrons)**.** *For two electrons, $q_1 q_2 = +e^2$, moving in parallel so that $\mathbf{v}_1 \cdot \mathbf{v}_2 = v^2 > 0$, the motional term is negative and reduces the repulsion. As $v \to c$ it cancels the static term exactly and the total force tends to zero.*

Proof. Immediate from (30): the bracket becomes $1 - v^2/c^2 \to 0$. □

The sign structure of (30) is the familiar one: like charges repel electrostatically and attract magnetically when their currents are parallel, and (29) is the sum. What the AEG reading adds is not the algebra — which is the standard tree-level Breit–Darwin coefficient — but the fact that the coefficient arises as a single pairing of one contact one-form with one inclined tangent, rather than as a sum of two separately postulated field contributions.

Warning 5.5 (Range of validity). Equation (29) is the quasi-static coefficient of order v^2/c^2; in electrodynamics it is the Darwin term of the two-body Lagrangian [6, 3]. It is not the exact force between two uniformly moving charges: the complete expression has further velocity-dependent terms of order v^4/c^4, retardation corrections, and spin–orbit contributions. Nothing in this section supplies those terms, and no all-orders statement is claimed. In the reading of Hypothesis 1.4 the truncation corresponds to the linearised contact flow (26), which is exact for constant θ but does not by itself generate the full Lorentz-covariant field.

5.4 Uniform motion in ripple geometry: compression without radiation

Paper 0, Section 4, reads the same word contravariantly as a family of pulled-back horocycles. For a prefix P_k, the ripple pencil is the family $\mathcal{R}(P_k) = \{P_k^{-1}(\mathcal{H}_t)\}$ of circles tangent at the pole $p_k = P_k^{-1}(\infty)$ (Theorem 4.2). The two cases now read as follows.

(i) *Charge at rest.* The pole is fixed, and the wavefronts of successive times are concentric or commonly tangent circles. The resulting field has the $1/r^2$ shape of a static configuration.

(ii) *Charge in uniform motion.* A Lorentz boost is a hyperbolic Möbius transformation, and the celestial sphere's Lorentz group is $\mathrm{PSL}_2(\mathbb{C})$, whose two-dimensional section is $\mathrm{PSL}_2(\mathbb{R})$ — the AEG Möbius group itself. Uniform motion therefore acts on the ripple pencil by a hyperbolic element $D_{e^\eta} \in H$, compressing the wavefronts ahead and stretching them behind, while leaving the pencil's common tangency point fixed: the two bivaluation data (q^+, q^-) are unchanged, so the history stays inside a single stabilizer orbit (Paper IV, Theorem 3.5 and Corollary 4.4, together with Proposition 4.6 for the normalizer).

The second case is the geometric form of the statement that a uniformly moving charge carries its near field with it and does not shed radiation: the pencil is compressed, not disrupted. Section 6 proves the corresponding algebraic statement — that a closed history with constant kernel coordinate has holonomy exactly 1 — and shows that the statement fails as soon as the kernel coordinate moves.

Remark 5.6 (The $v \to c$ limit in the $\mathfrak{E}_0$ picture). Corollary 5.4 can be read in $\mathfrak{E}_0$ as follows. As the common tilt of two histories approaches the critical rate, their tangent directions approach a common asymptotic slope, i.e. the two curves approach a common ideal point. In the picture of Paper 0, Theorem 3.9, the relative rate of shear between them decreases like $1 - \beta^2$, so the relative transverse disruption of the two pencil families vanishes. This is a heuristic reading of the coefficient, not a derivation of it: the derivation is Proposition 5.3, and the heuristic is included because it is what makes the coefficient visible inside $\mathfrak{E}_0$.

6 Accelerated Charges: Moving Poles and Projective Radiation

本节处理「加速电荷为什么会辐射」。AEG 的答案出奇地干净：匀速运动停留在固定核坐标 b 的稳定子轨道上，绕任何闭回路的乘性和乐 ** 望远镜抵消为 1** （这就是「无辐射定理」）；一旦发生加速，核坐标（涟漪极点）跳动 $b_1 \to b_2 \neq b_1$，四步闭环上的交叉比和乐 $\Lambda(\square) = \dfrac{(1+a_2b_1)(1+a_1b_2)}{(1+a_1b_1)(1+a_2b_2)}$ 就不再等于 1——而且它既不依赖标架截面，也无法被望远镜抵消。本节还给出 $\mathbb{F}_7$ 上的精确见证 $\Lambda = 4$，并说明这正是 Paper IV 定义 4.10 / 定理 4.11 所记录的对象。

6.1 Ripple pencils with a moving tangency point

Paper 0, Subsection 4.3, records the geometric state that becomes available once the second slot is read projectively:

> "Thus a general right-expansion history should not be pictured as one fixed family of concentric circles. *Each prefix has its own ripple pencil, and the common tangency point of that pencil may move with k.* The appropriate geometric state is $(G_k, G_k(z_0), p_k, \mathcal{R}_k)$: the operator, the selected point, the pole, and the pulled-back wavefront family."

The four-tuple is the whole content of the radiation phenomenon in this framework: the operator carries the cumulative transformation, the selected point carries the current value, the pole carries where the wavefronts are tangent, and the pencil carries the wavefronts themselves. A history in which p_k is constant emits nothing new; a history in which p_k jumps emits a mismatch between two pencils.

6.2 Bivaluations and the comoving frame

Fix $P_k = \begin{pmatrix} A_k & B_k \\ C_k & D_k \end{pmatrix} \in \mathrm{PGL}_2(\mathbb{R})$ with $\Delta_k = A_kD_k - B_kC_k > 0$, acting on the reference pair $(q_0^+, q_0^-) = (0, \infty)$. The image and kernel directions are

$$q_k^+ = P_k(0) = \frac{B_k}{D_k} = -a_k, \qquad q_k^- = P_k(\infty) = \frac{A_k}{C_k} = \frac{1}{b_k}, \tag{31}$$

and the pole of the contravariant pullback is $p_k = P_k^{-1}(\infty) = -D_k/C_k$ (Paper 0, Theorem 4.2 and Section 5.1). The pair (a_k, b_k) is a regular bivaluation in the sense of Paper IV, Definition 3.1, provided $1 + a_kb_k \neq 0$.

Theorem 6.1 (Bivaluation–projector correspondence, [2, Thm. 3.2])**.** *There is a natural bijection between regular bivaluations and rank-one idempotents of K^2. In the affine chart with $v_a = \binom{-a}{1}$ and $\varphi_b = (-b \;\; 1)$,*

$$\Pi_{a,b} = \frac{v_a \otimes \varphi_b}{\varphi_b(v_a)} = \frac{1}{1+ab}\begin{pmatrix} ab & -a \\ -b & 1 \end{pmatrix}, \qquad \varphi_b(v_a) = 1 + ab. \tag{32}$$

The projector $\Pi_k = \Pi_{a_k,b_k}$ is the intrinsic object behind the "comoving frame" of the physical reading: it splits the value space into the image line L_k^+ (the longitudinal near field) and the kernel line L_k^- (the transverse wavefront direction), and it is the unique such splitting determined by the history's prefix. Everything in this section is a statement about products of these projectors.

6.3 Uniform motion: no radiation

Theorem 6.2 (Fixed-kernel flatness)**.** *Let a history have constant kernel coordinate $b_k \equiv b_0$ for all prefixes, and let the image coordinate return to its initial value, $a_m = a_0$. Then the multiplicative holonomy*

$$\lambda(e_k) := \frac{1 + a_k b_{k-1}}{1 + a_{k-1} b_{k-1}} \tag{33}$$

satisfies $\prod_{k=1}^{m} \lambda(e_k) = 1$. In particular no closed history with constant kernel coordinate carries a non-unit multiplicative holonomy.

Proof. With $b_{k-1} = b_0$ the product is

$$\prod_{k=1}^{m} \frac{1 + a_k b_0}{1 + a_{k-1} b_0} = \frac{1 + a_1 b_0}{1 + a_0 b_0} \cdot \frac{1 + a_2 b_0}{1 + a_1 b_0} \cdots \frac{1 + a_0 b_0}{1 + a_{m-1} b_0} = 1,$$

each numerator cancelling the next denominator, the last against the first because $a_m = a_0$. □

The theorem is the algebraic content of the second reading of Section 5.4: uniform motion keeps the ripple pencil's tangency point fixed, and a fixed tangency point gives no holonomy. By Hypothesis 1.5 this is the no-radiation statement.

6.4 Accelerated motion: the pole jumps

Now let the history include the inverting element $J(z) = -1/z$, i.e. a second-slot division. Then the prefix leaves the affine Borel sector and enters the other Bruhat cell, and the kernel coordinate becomes nonzero. A physical acceleration is a change of the kernel coordinate between one step and the next.

Theorem 6.3 (Cross-ratio holonomy)**.** *Consider the closed four-step base loop*

$$\square : \quad (a_1, b_1) \to (a_2, b_1) \to (a_2, b_2) \to (a_1, b_2) \to (a_1, b_1),$$

in which the image coordinate moves with the kernel held fixed, then the kernel moves with the image held fixed, and then both return. Assume every vertex and every step is transverse, i.e. all of $1 + a_i b_j$ and the two cross terms $1 + a_2 b_1$, $1 + a_1 b_2$ are nonzero. Then the multiplicative holonomy of the loop is

$$\Lambda(\square) = \frac{(1 + a_2 b_1)(1 + a_1 b_2)}{(1 + a_1 b_1)(1 + a_2 b_2)} = 1 - \frac{(a_2 - a_1)(b_2 - b_1)}{(1 + a_1 b_1)(1 + a_2 b_2)}. \tag{34}$$

It is invariant under changes of the frame lifts Φ_k, and

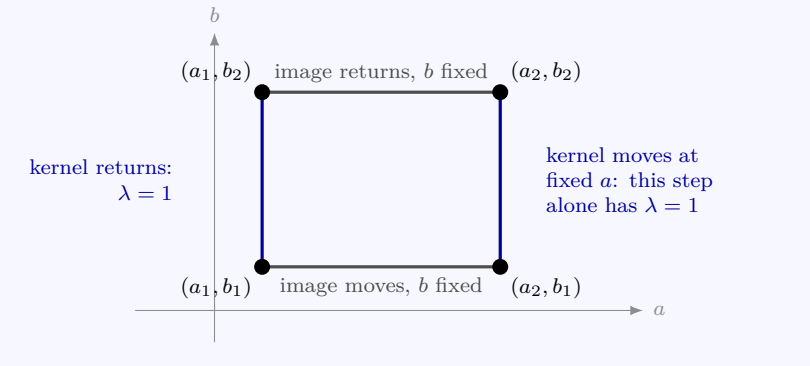
$$\Lambda(\square) \neq 1 \quad \Longleftrightarrow \quad a_2 \neq a_1 \text{ and } b_2 \neq b_1. \tag{35}$$

Moreover, as an identity between projectors,

$$\Pi_{a_1,b_1} \Pi_{a_1,b_2} \Pi_{a_2,b_2} \Pi_{a_2,b_1} = \Lambda(\square)^{-1} \, \Pi_{a_1,b_1}, \tag{36}$$

so $\Lambda(\square)$ is a gauge-invariant observable of the four idempotents alone.

Proof. Along $e_1 : (a_1, b_1) \to (a_2, b_1)$ and $e_3 : (a_2, b_2) \to (a_1, b_2)$ the factor (33) equals $(1 + a_2 b_1)/(1 + a_1 b_1)$ and $(1 + a_1 b_2)/(1 + a_2 b_2)$ respectively, because the kernel coordinate is the *previous* one in each case. Along e_2 and e_4 the image coordinate is constant, so $(1 + a_k b_{k-1})/(1 + a_{k-1} b_{k-1}) = 1$.



$$\Lambda(\square) = \frac{(1 + a_2 b_1)(1 + a_1 b_2)}{(1 + a_1 b_1)(1 + a_2 b_2)} \neq 1 \quad \text{iff both } a \text{ and } b \text{ move}$$

Figure 3: The four-step bivaluation square. Each individual step has a corrected base factor, but only the two image-moving steps contribute; the product is the cross-ratio (34). A loop in which the kernel coordinate never moves collapses to $\Lambda = 1$ by Theorem 6.2.

The product is (34). Gauge invariance: a change $\Phi_k \mapsto c_k \Phi_k$ multiplies each $\lambda(e_k)$ by c_k / c_{k-1}, whose product over a closed loop is 1. For (35), subtract numerator from denominator of (34): the difference is $(a_2 - a_1)(b_2 - b_1)$, which is nonzero exactly when both coordinates move. Identity (36) is a direct product of the rank-one matrices (32); it is proved symbolically in Appendix A and checked exhaustively over $\mathbb{F}_7$ there. □

Corollary 6.4 (Larmor criterion, algebraised)**.** *A closed history radiates, in the sense of Hypothesis 1.5, precisely when both the image coordinate and the kernel coordinate move. In the infinitesimal limit the log-holonomy is*

$$\log \Lambda(\square) = -\frac{(a_2 - a_1)(b_2 - b_1)}{(1 + a_1 b_1)(1 + a_2 b_2)} + O(3) = -\iint_{\square} \frac{\mathrm{d}a \wedge \mathrm{d}b}{(1 + ab)^2} + O(3), \tag{37}$$

so that displacement alone and velocity change alone are both non-radiative, while their product is radiative.

Proof. The first equality is the logarithm of (34) expanded with $(a_2 - a_1)(b_2 - b_1)$ small. The second is the same expansion written as an integral over the coordinate rectangle; Check V3.1 verifies it symbolically. The vanishing statement is (35). □

Remark 6.5 (The structure of the answer). The criterion has the shape of the classical radiation statement — a charge at rest radiates nothing, a uniformly moving charge radiates nothing, and an accelerated one does — but the two "nothing" cases are proved by different mechanisms: the rest case has $da = 0$ and the uniform case has $db = 0$. The criterion is therefore not "velocity change causes radiation" but "*simultaneous* displacement and change of the wavefront tangency point causes radiation", which in the affine sector reduces to the classical statement.

6.5 An exact finite witness

Example 6.6 ($\mathbb{F}_7$ calibration)**.** Let $K = \mathbb{F}_7$, so that $|G| = 336$, $|\operatorname{Biv}| = 56$ and $|H| = 6$ by Paper IV, Proposition 3.7 and Example 3.8. Take the loop

$$(a_1, b_1) = (0, 0), \quad (a_2, b_1) = (1, 0), \quad (a_2, b_2) = (1, 1), \quad (a_1, b_2) = (0, 1).$$

All four vertices are transverse, since $1 + a_i b_j \in \{1, 2\} \subset \mathbb{F}_7^\times$. Then

$$\Lambda(\square) \equiv \frac{(1+0)(1+0)}{(1+0)(1+1)} \equiv 2^{-1} \equiv 4 \pmod 7,$$

so $\Lambda(\square) = 4 \neq 1$ in $H \cong \mathbb{F}_7^\times$.

This is the exact finite witness that the obstruction is not an artefact of the real or complex topology: the holonomy is computed in a finite group of order 336, and it is a nonidentity element of the fiber H. Appendix A records the same example together with the exhaustive check of (36) over all transverse loops in $\mathbb{F}_7$.

6.6 The physical dictionary, stated as a dictionary

Warning 6.7 (What Hypothesis 1.5 adds). Theorem 6.2 and Theorem 6.3 are theorems about idempotents. They say: a closed history whose kernel coordinate is constant has a flat multiplicative transport, and a closed history that moves both coordinates has a holonomy with an exact cross-ratio value. They do not by themselves say that anything is emitted. The identification of $\Lambda(\square) \neq 1$ with radiation is Hypothesis 1.5, and the identification of its magnitude with radiated power is a further, unproved refinement.

Three dictionary readings are available, and each should be judged on its own:

(a) *Wavefront dislocation.* The two pencils $\mathcal{R}_k$ and $\mathcal{R}_{k+1}$ are tangent at different ideal points, so at their interface the wavefronts cannot nest smoothly; the mismatch is a caustic, i.e. a kink in the propagating wavefront. The magnitude of the mismatch is $|p_{k+1} - p_k|$ together with the holonomy Λ.

(b) *Field falloff.* In electrodynamics the field of an accelerated source has a term falling off as $1/r$ and proportional to the transverse acceleration, while the fields of static and uniformly moving sources fall off faster. The correspondence asserted by Hypothesis 1.5 is that the non-telescoping holonomy is the AEG carrier of that $1/r$ term. No fall-off exponent is derived here.

(c) *Phase residue.* By Remark 2.9 the loop's non-closure along the Reeb fiber measures a phase; a phase that does not return to itself is the same kind of statement as a radiated quantum's escape.

6.7 Relation to the AEG manuscripts

The construction proved in this section is not new to the AEG series: it is Paper IV, Definition 4.10 and Theorem 4.11, with the $\mathbb{F}_7$ witness of Example 4.13. It was developed in the dialogue summarised here and recorded in the manuscript under the heading of the projective-torsion problem. The present section's contribution is therefore not the theorem but the reading: the non-telescoping holonomy of Paper IV, Theorem 4.11 is the AEG object that the physical narrative calls "radiation", and Corollary 6.4 states the Larmor criterion in its terms.

Section 7 completes the picture by packaging the same data into a transport that also reproduces the affine torsion of Paper 0 on the Borel sector.

7 The Projective Torsion Transport

上一节的 $\Lambda(\square)$ 是个「闭回路可观测量」，但它只是和乐，还不是一个 联络。AEG 手稿里的原始提法是：要一个规范等变的比较量，限制到 Borel 扇区 B_∞ 时**退化回** Paper 0 的仿射平移挠率，而在闭回路上又**不能被望远镜抵消**。本节把这两个要求在同一个对象里同时满足：把每条历史边的算子步提升为半直积群 $\mathrm{Aff}(1,K) = K^\times \ltimes K$ 中的 **射影挠率传递元** $\mathcal{T}(e) = (\Lambda(e), \Delta\hat{\xi}(e))$。两个分量一个管乘性曲率、一个管加性挠率，限制到 $b \equiv 0$ 时前者归 1、后者一字不差地退回 $\hat{\xi}_n$。本节同时指出附带的规范化因子 2 问题（附录 B）。

7.1 The problem, as AEG states it

Paper IV, Section 13, lists as its first open problem:

> **Projective torsion.** Construct a gauge-equivariant comparison on $G \to G/H$ that reduces to affine translation torsion in the Borel sector and has a nontrivial closed-loop observable not forced to telescope [2].

Paper IV, Proposition 4.9, explains why the naive attempt fails: if the edge label of a lifted step is defined as the ratio $r_t = h_t h_{t-1}^{-1}$ of vertex frame lifts, then along every closed lifted path $r_t r_{t-1} \cdots r_{s+1} = h_t h_s^{-1}$ telescopes pairwise, so every closed path has trivial product. A nontrivial holonomy therefore requires the edge datum to couple the base motion to the previous frame, not just to record the frames.

7.2 The incidence-coupled transport

The needed coupling is supplied by the projectors of Theorem 6.1 without any additional structure: the natural pairing between a new image point and the preceding kernel covector is $\varphi_{b_{k-1}}(v_{a_k}) = 1 + a_k b_{k-1}$.

Definition 7.1 (Projective torsion transport, [2, Def. 4.10])**.** Let K be a field, let $e_k : g_{k-1} \to g_k$ be a lifted step between regular bivaluations with frame coordinates

$$g_k = \begin{pmatrix} 1 & -a_k \\ -b_k & 1 \end{pmatrix} \begin{pmatrix} \Phi_k & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Phi_k \in K^\times \cong H, \tag{38}$$

and suppose the step is transverse, $1 + a_k b_{k-1} \neq 0$. The projective torsion transport of e_k is the element of the affine semidirect product $\mathrm{Aff}(1,K) = K^\times \ltimes K$ given by

$$\mathcal{T}(e_k) := \begin{pmatrix} \Lambda_k & \Delta\hat{\xi}_k \\ 0 & 1 \end{pmatrix} \in \mathrm{Aff}(1,K), \tag{39}$$

with multiplicative and additive components

$$\Lambda_k = \frac{\Phi_k}{\Phi_{k-1}} \cdot \frac{1 + a_k b_{k-1}}{1 + a_{k-1} b_{k-1}} \in K^\times, \qquad \Delta\hat{\xi}_k = -\Phi_{k-1}^{-1} \frac{a_k - a_{k-1}}{1 + a_{k-1} b_{k-1}} \in K. \tag{40}$$

The transport of a history $\gamma = (s_1, \ldots, s_n)$ is the chronological product $\mathcal{T}(\gamma) = \mathcal{T}(e_n) \cdots \mathcal{T}(e_1)$.

Two remarks on the definition. The first component generalises the base factor (33) of Section 6 by multiplying it with the scale ratio of the step; the second projects the image displacement onto the previous kernel covector and normalises by the frame scale. Both are forced once one asks that the pair $(\Lambda, \Delta\hat{\xi})$ be multiplicative along a history and $\mathrm{Aff}(1,K)$-valued.

7.3 The Borel reduction

Theorem 7.2 (Two checks, [2, Thm. 4.11])**.** *Let $\mathcal{T}$ be as in Definition 7.1.*

(a) (Exact Borel reduction.) *On the affine sector B_∞, where $b_k \equiv 0$ and $a_k = -\xi_k$ for every prefix $g_k(z) = \Phi_k z + \xi_k$, the base correction is identically* 1 *and*

$$\Lambda_k\big|_{B_\infty} = \frac{\Phi_k}{\Phi_{k-1}} = s_k, \qquad \Delta\hat{\xi}_k\big|_{B_\infty} = \Phi_{k-1}^{-1}(\xi_k - \xi_{k-1}). \tag{41}$$

Multiplying the matrices chronologically gives upper-left entry Φ_n and upper-right entry $\Phi_n \sum_k \Phi_k^{-1}(\xi_k - \xi_{k-1}) = \xi_n$, which is the affine cocycle law of Paper 0, Proposition 7.1. In particular, for two scale-compatible histories the translation component of $\mathcal{T}(\delta)^{-1}\mathcal{T}(\gamma)$ is exactly the relative affine defect $\Phi^{-1}(\xi_\gamma - \xi_\delta)$ of Paper 0, Section 7.4.

(b) (Non-telescoping holonomy off the sector.) *On the four-step loop $\square$ of Theorem 6.3 with closed frame lift $\Phi_4 = \Phi_0$, the multiplicative component of $\mathcal{T}$ reproduces $\Lambda(\square)$ of (34). Hence the transport has a closed-loop observable that is gauge-invariant and is not forced to telescope.*

(c) (Modulated additive torsion.) *Off B_∞ the additive component carries the dynamic pole in its denominator, so the accumulated translation is a projective torsion modulated by the moving ripple pole rather than the affine one.*

Proof. (a) Put $b_k = 0$ and $a_k = -\xi_k$ in (40): the factor $(1 + a_k \cdot 0)/(1 + a_{k-1} \cdot 0)$ is 1, leaving $\Lambda_k = \Phi_k/\Phi_{k-1}$ and $\Delta\hat{\xi}_k = -\Phi_{k-1}^{-1}(-\xi_k + \xi_{k-1}) = \Phi_{k-1}^{-1}(\xi_k - \xi_{k-1})$. The matrix product of the $\mathrm{Aff}(1, K)$-elements is $\begin{pmatrix} \Phi_n & \Phi_n \sum \Phi_k^{-1}(\xi_k - \xi_{k-1}) \\ 0 & 1 \end{pmatrix}$, and the sum telescopes to ξ_n because $\xi_0 = 0$ (Paper 0, Proposition 7.1). (b) is Theorem 6.3, whose gauge invariance was proved there. (c) is the presence of $1 + a_{k-1}b_{k-1}$ in the denominator of $\Delta\hat{\xi}_k$, which is not a function of the charges alone once $b \neq 0$. $\square$

Remark 7.3 (Why the two demands can be met at once)*.* The Borel reduction of part (a) and the non-telescoping holonomy of part (b) look like competing requirements, because a quantity that is multiplicative and vanishes on $b \equiv 0$ might be expected to vanish identically. They coexist because they live in different components of $\mathrm{Aff}(1, K)$: the reduction uses the additive component (the translation $\Delta\hat{\xi}$, which becomes the affine cocycle), while the obstruction uses the multiplicative component (the cross ratio). On the Borel sector the multiplicative component degenerates to the step's own scale ratio, which telescopes as it must; off it, the same component carries the extra base factor and does not.

7.4 The curvature of the transport

For $K = \mathbb{R}$ or $\mathbb{C}$ one can pass to the infinitesimal picture. Paper IV, Remark 4.12, records the reductive connection whose solder component and connection one-form are

$$\omega_H = \Phi^{-1}\mathrm{d}\Phi + \frac{a\,\mathrm{d}b - b\,\mathrm{d}a}{1 + ab}, \qquad \theta_+ = \frac{\Phi^{-1}\mathrm{d}a}{1 + ab}, \tag{42}$$

and it records two facts about them that were stated loosely in an earlier version of that remark and are now exact. First, its curvature is $2(1+ab)^{-2}\mathrm{d}a \wedge \mathrm{d}b$, twice the para-Kähler form; the first-order transport density of the cross-ratio holonomy is that form with a minus sign, so the transport is the

holonomy of the connection whose transverse part is halved (Paper IV, Proposition 4.17). Second, on the Borel sector the solder component evaluates to

$$\theta_+\big|_{b=0} = \Phi^{-1}\mathrm{d}a = -e^{-M}\mathrm{d}\xi, \tag{43}$$

because the image coordinate satisfies $a = -\xi$ there; this is *minus* the left Maurer–Cartan translation form $e^{-M}\mathrm{d}\xi$ of Paper 0, Proposition 7.4, Equation (60). The minus sign is the sign of $a = -\xi$, not a different convention for the connection.

The comparison of the two densities is Lemma 7.4.

Lemma 7.4 (Two densities, one factor)**.** *The curvature of the connection* (42) *is* $\mathrm{d}\omega_H = 2(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$*, whereas the first-order log-holonomy density of Corollary 6.4 is* $-(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$*. The two differ by the factor* -2 *and by the choice of orientation.*

Proof. Since the Φ-term of ω_H is closed, only $\omega_2 = (1+ab)^{-1}(a\,\mathrm{d}b - b\,\mathrm{d}a)$ contributes. With $f = (1+ab)^{-1}$ one has $\mathrm{d}\omega_2 = (\partial_a(fa) + \partial_b(fb))\mathrm{d}a\wedge\mathrm{d}b$, and $\partial_a(fa) + \partial_b(fb) = -2ab(1+ab)^{-2} + 2(1+ab)^{-1} = 2(1+ab)^{-2}$. The transport density is Corollary 6.4. Check V4.1 verifies the coefficient symbolically. □

Remark 7.5 (Status of the factor)*.* Lemma 7.4 is a normalisation observation, not a contradiction: the cross-ratio formula (34) is exact and verified, and the curvature statement of the manuscript is exact as a statement about (42). The two normalisations differ by 2 because the transport of Definition 7.1 is the holonomy of $\frac{1}{2}$ of the connection's solder–connection split, i.e. of a connection whose second term carries a factor $\frac{1}{2}$. Any statement that identifies the radiated power with a curvature integral must therefore fix this normalisation first. Appendix B records the computation.

A second observation, unrelated to the factor: three cross-references inside Paper IV, Subsection 4.10–4.13 (`ex:f7-counts`, `prop:telescoping`, `eq:chart-projector`) resolved to `??` in the build of 20 October 2026, because those labels did not exist. They have since been repaired in the manuscript source — the example of Subsection 3.6 is now labelled, and the other two references point to Proposition 4.9 and to the projector display (32) — and the rebuilt Paper IV reports no undefined reference. Appendix D records the repair together with a second, smaller one in Paper 0.

7.5 The finite witness, restated

Example 6.6 applies verbatim to the transport: on $K = \mathbb{F}_7$ with the loop $\square$ taking $(0,0)\to(1,0)\to(1,1)\to(0,1)\to(0,0)$ and $\Phi_k \equiv 1$, one gets $\Lambda(\square) = 4 = 2^{-1} \in \mathbb{F}_7^\times$, a nonidentity element of H. With $\Phi_k \equiv 1$ the additive components are $\Delta\hat{\xi}_k = -(a_k - a_{k-1})/(1 + a_{k-1}b_{k-1})$, so the transport is a purely finite, exactly computable object. Appendix A lists the four edge transports of this loop explicitly.

7.6 Where this leaves the open problem

Theorem 7.2 supplies both demands of the problem statement: the transport reduces exactly to affine torsion on B_∞, and off B_∞ it has a closed-loop observable that is gauge-invariant and provably not forced to telescope, with a finite witness. What the theorem does not supply is a physical calibration of the transport's *units*: no statement is made about which physical quantity $\Lambda(\square) - 1$ measures, beyond the dictionary of Hypothesis 1.5. That is the open end recorded in Section 10.

8 Dynamical Scattering: Møller and Bhabha Sectors

两个带电粒子「带着动量碰撞」在 AEG 里对应什么？本节给出一个纯代数的区分：e^-e^- 散射（Møller）只有 t-道与 u-道，两者都留在仿射扇区 B_∞ 内；而 e^+e^- 散射（Bhabha）多出一条 s-道，它在 AEG 中恰好是**跨越 Bruhat 大胞腔**：$\mathrm{PGL}_2 = B_\infty \sqcup B_\infty J B_\infty$（Paper 0 公式 47），必经一次反演 $J(z) = -1/z$。总加法荷归零（$+e - e = 0$）正是「分母归零」的代数来源，而射影延拓让人从 $z = 0$ 穿到 ∞。这里要提醒：这是**结构对应**，不是振幅的推导。

8.1 Two electrons: both channels inside the affine sector

Let both particles be electrons. Then the total additive charge of any history containing both is $-2e \neq 0$, so no closed charge loop and hence no annihilation channel exists: by Proposition 4.5 the closed-loop mechanism requires opposite charges.

Two histories remain, and both are affine.

(i) *Threading order.* The word exchanges one quantum while the two world lines keep their order in the scale coordinate; this is the "t-channel" history of the exchange, and it is a word in B_∞ whose total charge is $(-e, -e)$.

(ii) *Crossed order.* The two final-state world lines are exchanged, so the appropriate comparison is between two histories γ and δ whose scale steps are interleaved; this is the "u-channel" of identical particles, and by Theorem 8.9 of Paper 0 the relative defect is the weighted area between the two charge chains.

In both cases the surviving velocity dependence is the one computed in Proposition 5.3: the two histories differ in their direction angle $\theta_1 - \theta_2$, and the pairing gives the coefficient $1 - \mathbf{v}_1 \cdot \mathbf{v}_2/c^2$ relative to the Coulomb term. For collinear motion this is the correction computed in Corollary 5.4.

Remark 8.1 (On spin). A spin-$\frac{1}{2}$ charge would need the double cover of the projective group, i.e. the passage from $\mathrm{PSL}_2(\mathbb{C})$ to $\mathrm{SL}_2(\mathbb{C})$, whose two-dimensional representation is generated by the Pauli matrices. Nothing in Paper 0 or Paper IV supplies that cover, and no spin-dependent statement is proved in this note. The remark is recorded because the physical narrative contains it.

8.2 Electron and positron: the second Bruhat cell

Now let the particles be an electron and a positron. Two of the three channels are affine, and one is not.

(i) *Exchange without annihilation.* The "t-channel" photon exchange is the configuration of Section 4: a bound-state history in B_∞ with potential $V = -\alpha_e/r$ and defect $e(1 - e^{-M_*})$ by Proposition 4.5.

(ii) *Annihilation.* At high relative momentum the pair acquires a channel with no like-charge analogue. Its algebraic signature is that the total additive charge vanishes, $A_{\mathrm{tot}} = (+e) + (-e) = 0$, exactly as in the closed loop of Proposition 4.5.

(iii) *Regeneration.* The word then continues from the vanishing denominator, and the projective continuation of Paper 0, Section 5.1 carries it through infinity and back to a finite pair.

The third item is where the projective cell is entered, and AEG has an exact statement about it.

Theorem 8.2 (Bruhat decomposition, [1, Eq. (47) and Thm. 6.2])**.** *For any field K,*

$$\mathrm{PGL}_2(K) = B_\infty \sqcup B_\infty J B_\infty, \tag{44}$$

and every $F \in \mathrm{PGL}_2(K)$ with nonzero lower-left entry admits the exact factorization

$$F = T_{A/C} \circ D_{(AD-BC)/C^2} \circ J \circ T_{D/C}, \tag{45}$$

i.e. translate, scale, invert, translate. The first cell consists of operators whose pole is infinity; the second consists of operators with a finite pole in the chosen affine chart.

Proof. Paper 0, Theorem 6.2 and its proof; the two cases $C = 0$ and $C \neq 0$ are $F = T_{B/D} \circ D_{A/D}$ and (45) respectively. □

The reading of the theorem in the scattering picture is now a translation of its two displayed statements: a history that stays in B_∞ has a pole at infinity, and a history that produces a finite pole must contain the inverting element exactly once.

process	charges	AEG sector	mechanism
$e^-e^- \to e^-e^-$	$(-e,-e)$, $A_{\mathrm{tot}} = -2e \neq 0$	B_∞ only	threading and crossed (identical-particle) orders; no closed charge loop
$e^+e^- \to e^+e^-$	$(-e,+e)$, $A_{\mathrm{tot}} = 0$	B_∞ and $B_\infty J B_\infty$	exchange, plus a channel that must cross J because the total additive charge vanishes
(e^-e^-) emission	$(-e,-e)$	B_∞, pole jumps	radiation as in Section 6: the pole may move inside the affine cell

8.3 What the correspondence does and does not assert

Warning 8.3 (Structural, not dynamical). The table above is a correspondence between channel types and Bruhat cells. It is not a derivation of a scattering amplitude, a cross section, or a Feynman rule for the annihilation channel. In particular:

(a) The vanishing of A_{tot} is an exact statement about charges (Theorem 8.6); the statement that a physical annihilation requires it is a dictionary statement.

(b) The projective continuation through an inadmissible division is exact in AEG (Paper 0, Section 5.1) but says nothing about which intermediate states are kinematically allowed. Ordinary division by zero remains inadmissible; what the continuation supplies is a value at the pole, not a physical process.

(c) The separation "t-channel inside B_∞, s-channel across $B_\infty J B_\infty$" is a two-cell statement. A quantitative Bhabha cross section would require a measure on histories, i.e. the path-integral structure of Remark 2.9, which is not constructed here.

What survives is the qualitative separation, and it is not empty: it explains why the like-charge and unlike-charge cases must differ in *kind* rather than only in sign, and it locates the difference in an exact algebraic invariant (the Bruhat cell of the composite) rather than in the sign of a coupling.

9 Bound States: Closed Loops and Bohr–Sommerfeld Quantization

电动力学的另一面是束缚态：正负电荷若既有吸引、又有切向运动，就既不湮灭也不逃逸，而是形成周期轨道。在 AEG 里这正是**椭圆型**不动点：连续分数迭代 $z_{k+1} = E - e^2/z_k$ 的判别式 $E^2 - 4e^2 < 0$ 时，两个不动点落在上半平面内部的共轭复点上，算子变成旋转。把这条闭合轨道水平提升到切触空间，它绕一圈不闭合，而是沿着 Reeb 纤维漂移 $\Delta A = -\oint a\, dM$。把这个漂移取成基本作用量的整数倍（Boothby–Wang 预量化），就得到玻尔–索末菲条件，代入库仑势直接给出 $E_n = -\dfrac{m\alpha_e^2}{2\hbar^2 n^2}$。本节末尾说明这一推导的**半经典**身份。

9.1 Hyperbolic, parabolic, and elliptic histories

A projective motion $F \in \mathrm{PSL}_2(\mathbb{R})$ is classified by its fixed-point set, and by Paper 0, Subsection 5.5, the finite fixed points solve

$$Cz^2 + (D - A)z - B = 0. \tag{46}$$

(i) *Hyperbolic*: two distinct real fixed points. The motion is a translation along the geodesic joining them; in the flow (26) the scale coordinate advances monotonically. A pair of charges in radial infall or escape follows this type.

(ii) *Parabolic*: one double fixed point, necessarily on the boundary. The corresponding motion is the horocyclic translation X_s of (1).

(iii) *Elliptic*: two complex conjugate fixed points in $\mathbb{C} \setminus \mathbb{R}$, i.e. inside $\mathbb{H}^2$ when read in the upper half-plane model. The motion is conjugate to a rotation, and its orbit is a closed curve around the interior fixed point. A bound two-body motion follows this type.

The Coulomb problem supplies the elliptic case through the continued-fraction map of Paper 0, Subsection 5.4. There, the stationary right-reciprocal update is $F(z) = a + b/z$ with matrix $M_F = \begin{pmatrix} a & b \\ 1 & 0 \end{pmatrix}$ and fixed points $z^2 - az - b = 0$. Writing the bound-state iteration as

$$z_{k+1} = E - \frac{e^2}{z_k}, \qquad M_F = \begin{pmatrix} E & -e^2 \\ 1 & 0 \end{pmatrix}, \tag{47}$$

the fixed-point equation becomes $z^2 - Ez + e^2 = 0$, whose discriminant is $\Delta = E^2 - 4e^2$. For $E < 2e$ the discriminant is negative, the two fixed points are complex conjugates, and the motion is elliptic.

Remark 9.1 (Status of the continued fraction). Equation (47) is the algebraic statement of Paper 0, Subsection 5.4 with $(a, b) = (E, -e^2)$. Its identification with the spectral problem of a Coulomb Hamiltonian is a dictionary statement, not a theorem: what AEG supplies is the conjugacy classification of the iteration, and what the dictionary supplies is the assertion that the physical bound states sit at the parameter values where that classification becomes elliptic.

9.2 The Reeb holonomy of a closed orbit is the action

Let γ_{orb} be a closed orbit of a bound motion in $\mathbb{H}^2_{(M,a)}$, and let $\tilde{\gamma}_{\mathrm{orb}}$ be its horizontal lift to $\mathcal{C}$. By Theorem 3.6(2) the lift does not close in $\mathcal{C}$: it ends at a Reeb height differing from its starting

height by

$$\Delta A(\gamma_{\mathrm{orb}}) = -\oint_{\gamma_{\mathrm{orb}}} a\,\mathrm{d}M = -\iint_{\Omega_{\mathrm{orb}}} \mathrm{d}a \wedge \mathrm{d}M, \tag{48}$$

where Ω_{orb} is the region enclosed by the orbit in the (M, a)-plane.

Lemma 9.2 (The action slot)**.** *Assume Hypothesis 1.2 and let the assignment slot of the contact form be filled by the scale-corrected radial momentum, $a := r\,p_r$. Then*

$$-\oint a\,\mathrm{d}M = -\oint p_r\,\mathrm{d}r, \tag{49}$$

so that the Reeb drift (48) is minus the classical mechanical action of the orbit.

Proof. $M = \log(r/\ell_0)$ gives $\mathrm{d}M = \mathrm{d}r/r$, hence $a\,\mathrm{d}M = rp_r\,\mathrm{d}r/r = p_r\,\mathrm{d}r$; the sign is the orientation of the loop. □

Warning 9.3 (What the substitution assumes)*.* Lemma 9.2 is the point at which the note departs from purely AEG statements: the assignment of the contact form is a number attached to a history, and calling it rp_r — with r the physical separation — is a dictionary step additional to Hypotheses 1.1–1.3. It is however the least avoidable one, because without it the Reeb drift is a dimensionless area and cannot be compared with $\hbar$.

9.3 Monodromy and the Bohr–Sommerfeld condition

Suppose the Reeb fiber is compactified. By the Boothby–Wang theorem [11, 10], if the contact form's Reeb field is a circle action with period $2\pi\hbar$, then $\mathcal{C}$ is the total space of a principal S^1-bundle — the prequantum bundle of the underlying symplectic manifold $\mathbb{H}^2$ — and the horizontal lift of a closed orbit closes in the fiber exactly when its drift is an integer multiple of the period.

Definition 9.4 (Monodromy condition)**.** The quantized contact model is the quotient $\mathcal{C}/(2\pi\hbar\,\mathbb{Z}\cdot R)$ of the Reeb flow. A closed orbit γ_{orb} defines a monodromy element $\exp(\mathrm{i}\,\Delta A(\gamma_{\mathrm{orb}})/\hbar) \in S^1$, and it defines a single-valued state exactly when

$$\oint_{\gamma_{\mathrm{orb}}} (-a\,\mathrm{d}M) = \oint p_r\,\mathrm{d}r = 2\pi n\hbar, \qquad n \in \mathbb{Z}. \tag{50}$$

Theorem 9.5 (Coulomb spectrum)**.** *Under Hypothesis 1.1–1.3, Lemma 9.2, and Definition 9.4, the bound orbits of a Coulomb pair in the radial coordinate satisfy*

$$\oint p_r\,\mathrm{d}r = 2\pi\Big(\frac{m\alpha_e}{\sqrt{-2mE}} - l\Big) = 2\pi n_r\hbar, \tag{51}$$

with $\alpha_e = e^2/(4\pi)$ and l the angular momentum. Writing $n = n_r + l/\hbar$, the energies are

$$E_n = -\frac{m\alpha_e^2}{2\hbar^2 n^2}, \qquad n = 1, 2, 3, \ldots \tag{52}$$

Proof. The second equality in (51) is the standard Coulomb radial action; Appendix C evaluates the half-action $\int_{r_-}^{r_+} p_r\,\mathrm{d}r = \pi(m\alpha_e/\sqrt{-2mE} - l)$ both in closed form and by numerical quadrature at three parameter points (Checks V7.1). Dividing by $2\pi\hbar$ and putting $n_r = n - l/\hbar$ gives $m\alpha_e/\sqrt{-2mE} = \hbar n$, hence $-2mE = m^2\alpha_e^2/(\hbar^2 n^2)$ and (52). Check V7.2 verifies the quantization condition and Check V7.3 the ground-state value. □

Remark 9.6 (The residue picture). The integral (51) is a contour integral between the two turning points, i.e. between the two real roots of the quadratic that Paper 0, Proposition 5.4 attaches to the iteration. Its closed form is the residue difference between the two ends of the contour. In the bivaluation language of Section 6 the two ends are the two punctures $(q^+, q^-) = (0, \infty)$ of the reference pair, so the same integral is a residue pairing at the two ends of the frame. This is a suggestive identification, not a proof; it is recorded because it is the natural continuation of the present picture.

9.4 How semiclassical is this?

Warning 9.7 (Range of the derivation). Theorem 9.5 is a Bohr–Sommerfeld (EBK) computation, not a solution of the Schrödinger equation. Three limitations should be stated.

(a) The condition (50) quantizes a single action integral; a genuine quantization would need the full polarization and the half-form (metaplectic) correction of geometric quantization. For the Coulomb problem those corrections are known to conspire so that the leading Bohr–Sommerfeld spectrum is exact, but that is a property of this particular potential, not a consequence of the present construction.

(b) The Langer-type correction $l \mapsto l + \frac{1}{2}$ changes the spectrum beyond leading order; no statement about it is made here.

(c) Nothing above defines a Hilbert space, an inner product, or an operator algebra. The Reeb circle of Definition 9.4 is a compactification hypothesis, and the multiplicities of the resulting levels are not counted.

What the section does establish is narrower and exact: the quantity that Bohr–Sommerfeld quantizes is, in this picture, the Reeb drift of a closed horizontal loop — an object that exists in AEG independently of any physical reading — and the Coulomb input enters only through Lemma 9.2 and the classical integral of Theorem 9.5.

10 Dictionary, Status Ledger, and Open Ends

本节把全篇收拢成三张表：**对照表**（物理概念 ↔ AEG 对象 ↔ 出处）、**状态账**（每条结论依赖哪几条字典假设）、以及**开放端**（哪些地方本稿只是对应、不是推导）。一句话总结：AEG 提供了「荷、标度、切触曲率、交叉比和乐、Reeb 漂移、径向作用量」这套确切的数学；物理电动力学要靠四条（严格说五条）字典假设才接得上，接上之后库仑、安培、辐射、束缚态四件事的**符号与结构**都能读出来，但**数值标定**（尤其是辐射功率与 $\hbar$ 的单位）仍是空的。

10.1 The dictionary

physical notion	AEG object	reference and status
asymptotic (bare) state	ideal point $q \in \partial_\infty \mathbb{H}^2$	Paper 0, §§3–4; dictionary
dressed state at scale r	ordinary point p with $y_p = e^{-M_*}$	Prop. 2.3; dictionary
scale evolution	multiplicative motion M_m, $M = \lambda v$	Prop. 3.6, Thm. 8.6
virtual quantum emission	additive motion A_t at depth M_i	Eq. (4); Hyp. 1.1

continued

physical notion	AEG object	reference and status
charge of a particle	sign of the additive step; positron = $\mathcal{R}$ conjugate	Lemma 4.1; Hyp. 1.1
sum over diagrams	the fiber $\rho^{-1}(g_{p_0\to p})$	Prop. 2.3; exact
bare coupling divergence	abscissa A_γ of the $\mathcal{A}$	Cor. 2.6; exact
renormalisation cancellation	vanishing weighted area $\tau(\gamma,\delta)$	Thm. 8.9; exact
path-integral phase	Reeb height $u_\gamma = A_\gamma/\mu$	Prop. 2.8, Rem. 2.9; exact object, dictionary reading
electrostatic potential	$a_1(r) = q_1 e^{-M_*}$	Eq. (18); Hyps. 1.1, 1.2
Coulomb sign rule	sign of $\kappa = \mu\lambda \propto q_1 q_2$	Eq. (20); Hyps. 1.1–1.3
binding energy of a dipole	$\tau(\ell,\varnothing) = e(1-e^{-M_*})$	Prop. 4.5; Hyp. 1.3
motion (velocity)	direction angle θ, tilt $\mathrm{d}M/\mathrm{d}\tau$	Thm. 3.9; Hyp. 1.4
Ampère force	one-form pairing $\alpha_1(\dot\gamma_2)$	Prop. 5.3; Hyps. 1.4, 1.3
frame compression by a boost	hyperbolic element $D_{e^\eta} \in H$ on the pencil	Thm. 4.2, Paper IV Thm. 3.5; exact object
Lorentz signature	indefinite Killing form of $\mathfrak{sl}_2$	Lemma 5.1; exact
no radiation of uniform motion	$\Lambda = 1$ for constant kernel coordinate	Thm. 6.2; exact, read via Hyp. 1.5
radiation of acceleration	$\Lambda(\square) \neq 1$ for moved kernel coordinate	Thm. 6.3; exact, read via Hyp. 1.5
sector of annihilation	Bruhat cell $B_\infty J B_\infty$	Thm. 8.2; exact object, dictionary reading
bound state	elliptic element; closed orbit inside $\mathbb{H}^2$	Prop. 5.4, §5.5; exact object
action of a bound orbit	Reeb drift $-\oint a\,\mathrm{d}M$	Thm. 3.6; Lemma 9.2
quantization	monodromy on the compactified Reeb circle	Def. 9.4; semiclassical

10.2 Status ledger

conclusion	hypotheses needed	confidence
the $p_0 \to p$ family is one operator fiber	none	theorem
the same family fans out in $\mathcal{A}$ with a weighted constraint	none	theorem
the two projections are Legendre-conjugate, with Δa versus ΔA	none	theorem (Paper 0, Thm. 9.9)
the kernel word realizes $p(1-e^{-q})$ three ways	none	theorem + check
κ changes sign exactly when one charge does	Hyp. 1.3	sign structure is exact; magnitude is dictionary
like charges repel, unlike attract	Hyps. 1.1–1.3	conditional; sign rule robust
the neutral pair has binding defect $e(1-e^{-M_*})$	Hyp. 1.3	conditional
$F = \frac{q_1 q_2}{4\pi r^2}\left(1 - \mathbf{v}_1\cdot\mathbf{v}_2/c^2\right)$	Hyps. 1.1–1.4	conditional; equals the Darwin coefficient

continued

conclusion	hypotheses needed	confidence
parallel currents attract the two electrons	same	conditional
uniform motion is non-radiative	Hyp. 1.5	algebraic part: theorem
acceleration produces a non-telescoping holonomy	Hyp. 1.5	algebraic part: theorem; with an $\mathbb{F}_7$ witness
radiated power $\propto \Lambda - 1$	Hyp. 1.5	*open*: no calibration
Møller inside B_∞, Bhabha across $B_\infty J B_\infty$	Hyps. 1.1, 1.3	correspondence only
Bohr–Sommerfeld spectrum $E_n = -m\alpha_e^2/(2\hbar^2 n^2)$	all five + Lemma 9.2	semiclassical, exact for Coulomb

10.3 What is exact and what is not

Three groups should be kept apart.

Exact AEG mathematics used here. The operator fiber of Proposition 2.3; the fan-out and the weighted constraint of Proposition 2.5; the Legendre conjugacy of Theorem 3.6 (Paper 0, Theorem 9.9); the three evaluations of the kernel word; the reflection conjugation of Lemma 4.1; the orientation reversal of Lemma 4.4 (Paper 0, Appendix D.1); the closed dipole loop (Paper 0, Appendix D.2); the no-radiation and cross-ratio theorems of Section 6 with the $\mathbb{F}_7$ witness; the Borel reduction and non-telescoping holonomy of Theorem 7.2 (Paper IV, Theorem 4.11); the Bruhat decomposition and factorization (Paper 0, Theorem 6.2); the Kepler conjugation classification (Paper 0, Proposition 5.4, Subsection 5.5); and the radial action computation of Appendix C.

Declared dictionaries. Hypotheses 1.1–1.4. Each names a physical quantity by an AEG object. They are not proved, not provable inside AEG, and not claimed.

Heuristics used for orientation only. (i) The "opening spring" and "annihilating composition" pictures of Section 4.5; (ii) the relative-shear reading of the $v \to c$ limit in Remark 5.6; (iii) the wavefront-caustic reading of the holonomy in Warning 6.7; (iv) the residue-pairing remark of Remark 9.6; (v) the spin remark of Remark 8.1. These are explanatory devices. They are marked in place, and no conclusion rests on them.

10.4 Open ends

Open Problem 10.1 (A three-dimensional carrier)**.** Every force law quoted here carries the factor 4π imported from the solid angle of three-space. The AEG carriers used are one-dimensional in the multiplicative direction and two-dimensional in the contact model. What is required is an AEG model whose multiplicative direction is genuinely three-dimensional, so that the $1/r$ kernel and the 4π normalisation are computed rather than inserted.

Open Problem 10.2 (Units of the radiation observable)**.** Theorem 6.3 produces a number $\Lambda(\square) - 1$ from a history, and Hypothesis 1.5 calls it radiation. No statement converts it into a power. A calibration would need either (a) a variational principle on histories whose second variation reproduces the Larmor rate, or (b) an identification of $\log \Lambda$ with a phase whose derivative is an energy flux. Neither is available. The factor-two normalisation of Lemma 7.4 must be settled first.

Open Problem 10.3 (A measure on the operator fiber)**.** Proposition 2.3 identifies the sum over diagrams with a fiber of an evaluation map, and Remark 2.9 identifies a candidate phase. A

genuine path integral needs a measure on the fiber that is invariant under the frame changes of Paper IV, Theorem 3.5. Whether the Reeb length $e^{-M}\mathrm{d}A$ of Paper 0, Proposition 8.2, is the right quasi-invariant measure is open.

Open Problem 10.4 (Spin)**.** Remark 8.1: the double cover $\mathrm{SL}_2(\mathbb{C}) \to \mathrm{PSL}_2(\mathbb{C})$ and its two-dimensional representation are not part of the AEG structure used here. A spin-dependent force or a spin-statistics statement would require extending the frame torsor of Paper IV to the metaplectic cover.

Open Problem 10.5 (Beyond the Darwin order)**.** Warning 5.5: the coefficient of Proposition 5.3 is the $O(v^2/c^2)$ term. Whether the contact picture can produce the full velocity dependence — and, if so, whether the indefinite structure of Lemma 5.1 is the mechanism — is open.

10.5 Closing remark

The five working notes that this document consolidates were an exploration, and the exploration found something worth stating precisely: the same order defect that AEG proves to be a weighted area appears, under two conjugate projections, as a tearing of the value and as a drift of the phase; the same projective holonomy that AEG proves to be non-telescoping distinguishes uniform from accelerated motion; and the same Reeb drift that AEG proves to be a hyperbolic area is the quantity that Bohr–Sommerfeld quantizes. Those three identifications are exact. What connects them to electrodynamics is a dictionary, and the dictionary is stated above in five lines so that it can be argued with.

A Exact Matrix Computations

本附录把正文用到的矩阵计算全部写开：投影算子的乘积恒等式、交叉比和乐、$\mathbb{F}_7$ 上的逐边传递元与总传递元，以及对 $\mathbb{F}_7$ 上全部横截闭环的穷举核对。所有条目对应复核脚本的 V1–V2 组。

A.1 The rank-one projectors

With $v_a = \binom{-a}{1}$ and $\varphi_b = (-b \;\; 1)$, the projector of Theorem 6.1 is

$$\Pi_{a,b} = \frac{v_a \otimes \varphi_b}{1+ab} = \frac{1}{1+ab}\begin{pmatrix} ab & -a \\ -b & 1 \end{pmatrix}, \qquad \Pi_{a,b}^2 = \Pi_{a,b}. \tag{53}$$

Two products are elementary and are used repeatedly:

$$\Pi_{a,b}\Pi_{a',b} = \Pi_{a,b} \quad (a' \text{ transverse}), \qquad \Pi_{a,b}\Pi_{a,b'} = \Pi_{a,b'}, \tag{54}$$

i.e. two projectors sharing a kernel covector compose to the first, and two sharing an image vector compose to the second. The pairing $\varphi_b(v_{a'}) = 1 + a'b$ cancels in the first product; it is exactly this pairing that survives, as a scalar, when neither the kernel nor the image is shared, and it is the edge factor (40) of the transport.

A.2 The four-step holonomy

Let $\square$ have vertices $(a_1,b_1),(a_2,b_1),(a_2,b_2),(a_1,b_2)$ traversed in that order. A direct product gives the identity

$$\Pi_{a_1,b_1}\,\Pi_{a_1,b_2}\,\Pi_{a_2,b_2}\,\Pi_{a_2,b_1} = \Lambda(\square)^{-1}\,\Pi_{a_1,b_1}, \qquad \Lambda(\square) = \frac{(1+a_2b_1)(1+a_1b_2)}{(1+a_1b_1)(1+a_2b_2)}. \tag{55}$$

Check V1.1 verifies (55) symbolically for arbitrary a_1, b_1, a_2, b_2 over a field of characteristic zero, and Check V1.2 verifies the equivalent form $1 - \frac{(a_2-a_1)(b_2-b_1)}{(1+a_1b_1)(1+a_2b_2)}$. Check V1.3 verifies that $\Lambda = 1$ identically when $a_2 = a_1$ or when $b_2 = b_1$.

A.3 Edge transports over the seven-element field

Take $K = \mathbb{F}_7$, all frame coordinates $\Phi_k = 1$, and the loop $(0,0) \to (1,0) \to (1,1) \to (0,1) \to (0,0)$. Definition 7.1 gives the following four edge transports, written as $(\Lambda_k, \Delta\hat{\xi}_k) \in \mathbb{F}_7^\times \times \mathbb{F}_7$ with $2^{-1} = 4$:

edge	from $\to$ to	Λ_k	$\Delta\hat{\xi}_k$	$\mathcal{T}(e_k) = \begin{pmatrix} \Lambda_k & \Delta\hat{\xi}_k \\ 0 & 1 \end{pmatrix}$
e_1	$(0,0) \to (1,0)$	1	$-1 \equiv 6$	$\begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix}$
e_2	$(1,0) \to (1,1)$	1	0	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
e_3	$(1,1) \to (0,1)$	$2^{-1} \equiv 4$	$2^{-1} \equiv 4$	$\begin{pmatrix} 4 & 4 \\ 0 & 1 \end{pmatrix}$
e_4	$(0,1) \to (0,0)$	1	0	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

The chronological product is

$$\mathcal{T}(\square) = \mathcal{T}(e_4)\mathcal{T}(e_3)\mathcal{T}(e_2)\mathcal{T}(e_1) = \begin{pmatrix} 4 & 4 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 6 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \tag{56}$$

since $4 \cdot 6 + 4 = 28 \equiv 0 \pmod 7$. Checks V2.4 and V2.5 verify the four edge entries and (56).

Remark A.1 (The additive component can cancel on a single loop). On this particular loop the additive component of the total transport vanishes identically, so the invariant is purely multiplicative. This is not a defect of the construction: the additive component is a sum of increments weighted by different denominators, and nothing forces it to survive. It does show that a single loop is not enough to exhibit both components at once; the loop of Appendix A.4 can be shifted to make $\Delta\hat{\xi}$ nonzero, and the Borel reduction of Theorem 7.2(a) is where the additive component is the whole story.

A.4 Exhaustive check over the seven-element field

For every ordered pair of transverse bivaluations $(a, b), (c, d) \in \mathbb{F}_7^2$ with all of $1 + ab$, $1 + cd$, $1 + cb$, $1 + ad$ nonzero, the identity (55) was verified entry by entry in pure modular arithmetic. There are 1447 such loops, and the identity holds for each (Check V2.3). The count 43 of transverse pairs in the affine chart (the remaining 13 of the 56 bivaluations of $\mathbb{P}^1 \times \mathbb{P}^1 \setminus \Delta$ lie on the line at infinity) is Check V8.4.

Warning A.2 (What the finite check does not prove). The exhaustive $\mathbb{F}_7$ check confirms the identity in characteristic 7 and confirms that the witness $\Lambda = 4$ is a genuine nonidentity holonomy. It does not prove the identity in characteristic zero — that is Check V1.1 — and it says nothing about the analytic statements of Sections 6–9, which are over $\mathbb{R}$.

B Curvature Normalization of the Projective Connection

本附录把「曲率形式」与「和乐密度」之间的因子 2 老老实实算一遍。结论是：交叉比和乐 $\Lambda(\square)$ 的一阶展开密度是 $-(1+ab)^{-2}\,\mathrm{d}a\wedge\mathrm{d}b$，而 Paper IV 备注 4.12 给出的约化联络 ω_H 的曲率是 $+2(1+ab)^{-2}\,\mathrm{d}a\wedge\mathrm{d}b$。两者都对，但**规范化不同**：传递元是那条联络" 焊接项取一半" 后的和乐。任何把辐射功率等同到曲率积分的说法，必须先把这个因子定下来。

B.1 The curvature of the reductive connection

Let $f=(1+ab)^{-1}$ and consider the non-closed part of the connection (42),

$$\omega_2 = f(a\,\mathrm{d}b - b\,\mathrm{d}a). \tag{57}$$

Because $\mathrm{d}(a\,\mathrm{d}b - b\,\mathrm{d}a) = 2\,\mathrm{d}a\wedge\mathrm{d}b$,

$$\mathrm{d}\omega_2 = \mathrm{d}f\wedge(a\,\mathrm{d}b - b\,\mathrm{d}a) + 2f\,\mathrm{d}a\wedge\mathrm{d}b. \tag{58}$$

Now $\mathrm{d}f = -(1+ab)^{-2}(b\,\mathrm{d}a + a\,\mathrm{d}b)$, so

$$\begin{aligned}\mathrm{d}f\wedge(a\,\mathrm{d}b - b\,\mathrm{d}a) &= -(1+ab)^{-2}\big[(b\,\mathrm{d}a + a\,\mathrm{d}b)\wedge(a\,\mathrm{d}b - b\,\mathrm{d}a)\big]\\ &= -(1+ab)^{-2}\cdot 2ab\,\mathrm{d}a\wedge\mathrm{d}b,\end{aligned} \tag{59}$$

since $(b\,\mathrm{d}a + a\,\mathrm{d}b)\wedge(a\,\mathrm{d}b - b\,\mathrm{d}a) = ab\,\mathrm{d}a\wedge\mathrm{d}b + ab\,\mathrm{d}a\wedge\mathrm{d}b$. Combining (58) and (59),

$$\mathrm{d}\omega_H = \mathrm{d}\omega_2 = 2\,\mathrm{d}a\wedge\mathrm{d}b\Big(\frac{1}{1+ab} - \frac{ab}{(1+ab)^2}\Big) = \frac{2\,\mathrm{d}a\wedge\mathrm{d}b}{(1+ab)^2}. \tag{60}$$

Check V4.1 verifies the coefficient $\partial_a(fa) + \partial_b(fb) = 2(1+ab)^{-2}$ symbolically.

B.2 The density of the transport

Independently, the first-order expansion of the cross-ratio holonomy around a coordinate rectangle of side lengths ε and η based at (a_0, b_0) is, by (34),

$$\log\Lambda(\square) = -\frac{\varepsilon\eta}{(1+a_0b_0)^2} + O(3) = -\iint_{\square}\frac{\mathrm{d}a\wedge\mathrm{d}b}{(1+ab)^2} + O(3), \tag{61}$$

which is Check V3.1. Comparing (60) with (61), the transport density is minus one half of the connection's curvature density.

B.3 Reconciliation

Three statements are exact and should be kept apart.

(i) The cross-ratio formula (34) is exact and is the holonomy of the idempotent transport of Definition 7.1 (Checks V1.1–V2.5).

(ii) The connection (42) has curvature $2(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$ (Check V4.1).

(iii) The first-order log-holonomy density is $-(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$ (Check V3.1).

The holonomy of a connection with curvature Ω around a small positively oriented loop is $\exp(-\int_\Sigma \Omega)$ in the abelian approximation. Statement (ii) would therefore give $\log\Lambda = -\int_\Sigma 2(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$, which differs from (iii) by the factor 2. The resolution is that the idempotent transport is the holonomy of the connection whose reductive second term is $\frac{1}{2}f(a\,\mathrm{d}b - b\,\mathrm{d}a)$, i.e. of $\omega_H^{(1/2)} := \Phi^{-1}\mathrm{d}\Phi + \frac{1}{2}f(a\,\mathrm{d}b - b\,\mathrm{d}a)$. Its curvature is $\frac{1}{2}\cdot 2(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$, whose negative integral is exactly (61).

Remark B.1 (Which normalisation is "right"). Neither is more correct. The factor is the difference between a connection normalised so that its transverse term is the full $(1+ab)^{-1}(a\,\mathrm{d}b - b\,\mathrm{d}a)$ and one normalised so that the transport's multiplicative component is the raw cross-ratio. What matters is that a physical identification of $\Lambda(\square)-1$ with a radiated power must state which convention it uses; otherwise the same history gives two answers differing by a factor two.

The two exact statements are now in Paper IV itself: Remark 4.12 carries the curvature identity $\mathrm{d}\omega_H = 2(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$ together with the correct Borel reduction of the solder component (equation (43) above), and Proposition 4.17 identifies the connection whose holonomy reproduces $\Lambda(\square)$ to first order. The choice between the two normalisations is left open there, as it is here.

Remark B.2 (Sign). The two densities also differ in sign, because the loop is traversed in the orientation that makes $\mathrm{d}a\wedge\mathrm{d}b$ positive while the log-holonomy is negative. Both statements are checked, and the signs are consistent with the holonomy formula's minus sign.

C The Coulomb Radial Action

本附录给出第 9 节用到的库仑径向作用量的完整初等推导，并对三个参数点做数值核对（V7.1）。结果是半作用量 $\int_{r_-}^{r_+} p_r\,\mathrm{d}r = \pi\big(m\alpha_e/\sqrt{-2mE} - l\big)$，整体作用量是它的两倍。由此立刻得到玻尔能谱。

C.1 The radial momentum and the turning points

For a Coulomb pair with potential $V = -k/r$ and conserved angular momentum l,

$$p_r^2 = 2m\Big(E + \frac{k}{r}\Big) - \frac{l^2}{r^2} = \frac{2mE\,r^2 + 2mk\,r - l^2}{r^2}. \tag{62}$$

For $E<0$ put $A = 2m|E| > 0$, so that $2mE = -A$. The turning points are the positive roots of $Ar^2 - 2mk\,r + l^2 = 0$:

$$r_\pm = \frac{mk \pm \sqrt{m^2k^2 - Al^2}}{A}, \qquad r_+ + r_- = \frac{2mk}{A}, \qquad r_+ r_- = \frac{l^2}{A}. \tag{63}$$

Between them $p_r = \sqrt{A}\,\sqrt{(r_+ - r)(r - r_-)}/r$.

C.2 The half-action

Lemma C.1 (Half radial action)**.** *With $A = 2m|E|$ and the turning points (63),*

$$\int_{r_-}^{r_+} p_r\,\mathrm{d}r = \pi\Big(\frac{mk}{\sqrt{A}} - l\Big) = \pi\Big(\frac{mk}{\sqrt{-2mE}} - l\Big). \tag{64}$$

Proof. Substituting $p_r = \sqrt{A}\sqrt{(r_+ - r)(r - r_-)}/r$ reduces the claim to the elementary integral

$$\int_a^b \frac{\sqrt{(b-x)(x-a)}}{x}\,\mathrm{d}x = \frac{\pi(a+b)}{2} - \pi\sqrt{ab}, \tag{65}$$

which is Check V7.4 at three parameter points. Applying it with $a = r_-$, $b = r_+$ and using (63),

$$\int_{r_-}^{r_+} p_r \,\mathrm{d}r = \sqrt{A}\,\pi\Big(\frac{r_+ + r_-}{2} - \sqrt{r_+ r_-}\Big) = \sqrt{A}\,\pi\Big(\frac{mk}{A} - \frac{l}{\sqrt{A}}\Big) = \pi\Big(\frac{mk}{\sqrt{A}} - l\Big).$$

The second form uses $A = -2mE$. □

Remark C.2 (On (65)). The integral is a standard contour integral; it can also be obtained by the Kepler substitution $x = \bar{r}(1 - e\cos\psi)$ with $\bar{r} = (a+b)/2$ and $e = (b-a)/(b+a)$. The version stated is the one verified numerically.

C.3 Numerical verification

Check V7.1 evaluates the left side of (64) by symbolic quadrature and compares with the closed form at three parameter points (m, k, l, E):

m	k	l	E	$\int_{r_-}^{r_+} p_r \,\mathrm{d}r$
1	1	1/2	$-1/2$	$1.57079632679\ldots = \pi/2$
2	3	1	$-9/8$	$5.74417322273\ldots$
1/2	3	1	$-8/9$	$1.85665065184\ldots$

In each case the quadrature agrees with $\pi(mk/\sqrt{-2mE} - l)$ to better than 10^{-12}.

C.4 The spectrum

With $\alpha_e = e^2/(4\pi)$ and $k = \alpha_e$, the full action is twice (64), and Quantization (50) gives

$$2\pi\Big(\frac{m\alpha_e}{\sqrt{-2mE}} - l\Big) = 2\pi n_r \hbar \quad\Longleftrightarrow\quad \frac{m\alpha_e}{\sqrt{-2mE}} = l + n_r\hbar = \hbar n, \tag{66}$$

where $n := n_r + l/\hbar$. Then

$$E_n = -\frac{m\alpha_e^2}{2\hbar^2 n^2}. \tag{67}$$

Check V7.2 substitutes (67) into the middle expression of (66) and finds $\hbar n$ identically; Check V7.3 recovers $E_1 = -m\alpha_e^2/(2\hbar^2)$.

Warning C.3 (Semiclassical status). As stated in Warning 9.7, (67) is a Bohr–Sommerfeld result. Its agreement with the exact hydrogen spectrum is a property of the Coulomb potential; no operator-theoretic statement is made, and the multiplicities n^2 of the levels are not derived.

D Register of Verified Identities

本附录是**可复核性**的部分。第一小节说明复核脚本怎么跑；第二小节把全部 29 项检查逐条列出；第三小节记录整理过程中对原始讨论所做的事实性更正（含一处代数系数、一处归一化因子、两处引用编号）；第四小节记录 AEG 手稿当前编译中发现的三个未解析引用与「开放问题已解却仍列在清单里」的情况。最后一小节说明哪些断言**没有**被复核。

D.1 The verification script

All checks below are produced by one script, kept beside this document as `verify-notes.py`, together with its recorded output `verification-report.txt`. It requires only `sympy` and `numpy` and is run with

`python3 verify-notes.py` (or with any interpreter that has `sympy` installed)

The script prints one `[PASS]`/`[FAIL]` line per check and exits non-zero if any check fails. On the last run, all 29 checks passed.

D.2 Register of verified identities

check	identity verified	where used
V1.1	$\Pi_{a_1,b_1}\Pi_{a_1,b_2}\Pi_{a_2,b_2}\Pi_{a_2,b_1} = \Lambda(\square)^{-1}\Pi_{a_1,b_1}$, symbolically in characteristic 0	Thm. 6.3, App. A.2
V1.2	$\Lambda = 1 - \frac{(a_2-a_1)(b_2-b_1)}{(1+a_1b_1)(1+a_2b_2)}$	(34)
V1.3	$\Lambda = 1$ identically if $a_2 = a_1$ or $b_2 = b_1$	(35)
V1.4	$\prod_k \frac{1+a_kb_0}{1+a_{k-1}b_0} = 1$ on a closed image loop	Thm. 6.2
V1.5a	$\Pi_{a,b}\Pi_{a',b} = \Pi_{a,b}$	App. A.1
V1.5b	$\Pi_{a,b}\Pi_{a,b'} = \Pi_{a,b'}$	App. A.1
V2.1	$\Lambda(\square) = 4$ over $\mathbb{F}_7$	Example 6.6
V2.2	$4 \neq 1$ in $\mathbb{F}_7^\times \cong H$	Example 6.6
V2.3	the projector identity over all 1447 transverse $\mathbb{F}_7$ loops	App. A.4
V2.4	the four edge transports $(1,6),(1,0),(4,4),(1,0)$	App. A.3
V2.5	$\mathcal{T}(\square) = \mathrm{diag}(4,1)$ in $\mathrm{Aff}(1,\mathbb{F}_7)$	(56)
V3.1	$\log\Lambda(\square) = -\frac{\varepsilon\eta}{(1+a_0b_0)^2} + O(3)$	Cor. 6.4, App. B.2
V4.1	$\mathrm{d}\omega_2 = 2(1+ab)^{-2}\mathrm{d}a\wedge\mathrm{d}b$	(60)
V4.2	curvature density / transport density $= 2$	Lemma 7.4
V4.3	$\theta_+\big\vert_{b=0} = \Phi^{-1}\mathrm{d}a = -e^{-M}\mathrm{d}\xi$	Thm. 7.2(a)
V5.1	$\alpha = -\mathrm{d}(aM - a + A) + M\,\mathrm{d}a$	Lemma 3.2
V5.2	the variant $-\mathrm{d}(A + aM - a) + (1+M)\mathrm{d}a$ is *not* α	Remark 3.3
V5.3	$e^{-M}\alpha = \mathrm{d}(e^{-M}a) - e^{-M}\mathrm{d}A$	Lemma 3.4
V6.1	Reeb drift $= p(1-e^{-q})$ by the charge route and by the area route	Lemma 3.8
V6.2	the same value is the additive charge of $h(p,q)$ (Paper 0, Thm. 8.6)	Lemma 3.8
V6.3	$\int_0^q\int_0^p e^{-M}\mathrm{d}A\,\mathrm{d}M = p(1-e^{-q})$	Lemma 3.8
V7.1	$\int_{r_-}^{r_+} p_r\mathrm{d}r = \pi\big(mk/\sqrt{-2mE} - l\big)$ at three parameter points	Lemma C.1
V7.2	$m\alpha/\sqrt{-2mE_n} = \hbar n$	(67)
V7.3	$E_1 = -m\alpha^2/(2\hbar^2)$	(67)
V7.4	$\int_a^b \frac{\sqrt{(b-x)(x-a)}}{x}\mathrm{d}x = \frac{\pi(a+b)}{2} - \pi\sqrt{ab}$	(65)
V8.1	$\lvert\mathrm{PGL}_2(\mathbb{F}_7)\rvert = 336$	Example 6.6
V8.2	$\lvert\mathrm{Biv}_{\mathbb{F}_7}\rvert = 56$	Example 6.6
V8.3	$\lvert H\rvert = 6$, all fibers of size 6	Example 6.6
V8.4	43 of the 56 bivaluations lie in the finite affine chart	App. A.4

D.3 Corrections made to the working notes

The five source documents were exploratory and contained several slips that a formal document cannot carry. They are corrected above and recorded here.

(a) **Legendre coefficient.** The note 勒让德形式`.md` writes $\alpha = \mathrm{d}a - \mathrm{d}A - (d(aM) - M\,\mathrm{d}a)$ and then displays $-\mathrm{d}(A + aM - a) + (1 + M)\mathrm{d}a$. The expansion gives $-\mathrm{d}(A + aM - a) + M\,\mathrm{d}a$; the coefficient is M, not $1 + M$. Corrected in Lemma 3.2 and Remark 3.3; verified negatively by Check V5.2. Nothing downstream depends on the slip, because every subsequent identity in that note uses the exponential form, which is correct.

(b) **Shell normalisation.** The note 电动力学讨论`-01.md` writes the shell contribution as $\mathrm{d}V = -\frac{\kappa}{4\pi}\mathrm{d}(e^{-M}) = +\frac{e^2}{4\pi r'^2}\mathrm{d}r'$. With $e^{-M} = \ell_0/r'$ the differentiated form carries a factor ℓ_0, and consistency with the stated $V = \frac{\kappa}{4\pi\ell_0}e^{-M}$ requires $\mathrm{d}V = \frac{\kappa}{4\pi\ell_0}\mathrm{d}(e^{-M})$. Corrected in (19)–(22).

(c) **Citation numbers.** Three citations in the notes were checked against the compiled manuscripts and corrected: the statement quoted as "Theorem 3.5" of Paper 0 is Proposition 3.5 (a Laplace eigenfunction statement); the "self-intersecting figure-eight charge path of Section 8.7" is not in Section 8.7 (which is *Tearing*); the relevant material is the four-step staircase of Subsection 8.5 and the singular-two-chain remark of Appendix D.2. Citations that were correct and are retained: Paper 0, Theorems 8.6, 8.9, 9.2; Propositions 9.1, 9.6, 9.8; Appendix D.1–D.2; equation (47) of Paper 0 for the Bruhat decomposition; Paper IV, Theorem 3.2, Corollary 4.4, Propositions 4.6 and 4.9, Example 3.8.

(d) **Numbering of the projective-torsion construction.** The notes refer to "Open Problem 14.1". In the compiled Paper IV the open problems are in Section 13, where the projective-torsion problem was its first entry before it was replaced by the resolution recorded in Appendix D.4(b); the construction itself is Definition 4.10, Theorem 4.11, Remark 4.12 and Example 4.13. Both are used with their current numbers.

D.4 Repairs made to the AEG manuscripts

Four defects were found while checking this note against the built manuscripts: three by auditing the citations of Section 1, and one in review of Section 7. The first three are not mathematical errors; the fourth is a genuine sign-and-factor error in a remark. All four have been repaired in the manuscript sources, in three separate commits, and are recorded here so that the repairs themselves remain auditable.

(a) **Three unresolved cross-references in Paper IV.** The build of 20 October 2026 reported `Reference `prop:telescoping', `eq:chart-projector' and `ex:f7-counts' undefined`. Repaired by labelling the example of Subsection 3.6 as `ex:f7-counts`, and by retargeting the other two references to `prop:exact-transport-telescope` (Proposition 4.9) and `eq:local-projector` (equation (6), the projector display of Subsection 3.1). A label `rem:reductive-connection` was added to Remark 4.12 so that it can be referred to at all.

(b) **An open problem already resolved in the main text.** Paper IV, Section 13, listed the projective-torsion problem as open, while Subsection 4.10–4.11 states and proves exactly that construction. Repaired by replacing the first entry of the open-problem list with a paragraph that records the resolution, cites Theorem 4.11 and Example 4.13, and keeps open only the normalisation question of Lemma 7.4. The governance record `governance/paper-IV-closure-report.md` carries a dated amendment to the same effect.

(c) **Four unresolved cross-references in Paper 0.** Found in the same pass: the Legendre-duality subsection of Paper 0, Section 9, referred to `prop:reeb-quotient`, `eq:contact-form`, `thm:charge-descent` and `prop:surface-never-closes`, none of which exists. Repaired by retargeting to `prop:p0-contact-quotient`, `eq:arithmetic-contact-form`, `thm:p0-descent` and `prop:p0-surface-does-not-close`. Note that this is the subsection that absorbed the Legendre material of Section 3: the same paste-in that carried the mathematics carried provisional label names.

(d) **Sign, factor, and a stale equation number in Paper IV, Remark 4.12.** Found in review of the present note. The remark defined $\theta_+ = \Phi^{-1}(1+ab)^{-1}\mathrm{d}a$ and then evaluated it on the Borel sector as $-\Phi^{-1}\mathrm{d}a$; with $a = -\xi$ the evaluation is $\theta_+|_{b=0} = \Phi^{-1}\mathrm{d}a = -e^{-M}\mathrm{d}\xi$, i.e. *minus* the left Maurer–Cartan form, and the minus sign belongs to the step, not to the definition. The same sentence identified the first-order expansion of $\Lambda(\Box)$ with $(1+ab)^{-2}\mathrm{d}a \wedge \mathrm{d}b$ without a factor, although the curvature of the connection displayed there is $2(1+ab)^{-2}\mathrm{d}a \wedge \mathrm{d}b$ and the transport density is $-(1+ab)^{-2}\mathrm{d}a \wedge \mathrm{d}b$ (the transport-density proposition of Paper IV, Subsection 4.8, and Lemma 7.4 above). The citation "Paper 0, Equation (58)" was also stale: the left Maurer–Cartan translation form is Equation (60) in the current build, and (58) is an unrelated formula. Repaired by rewriting the remark: it now carries the curvature computation $\mathrm{d}\omega_H = 2(1+ab)^{-2}\mathrm{d}a \wedge \mathrm{d}b$ with its two-line derivation, the correct Borel evaluation (equation (43) above) with the reason for the sign, and a forward reference to Subsection 4.8 for the density comparison. Nothing is left in the remark as an unproved claim, and the factor is no longer described as an identification.

After the repairs, the builds of 20 October 2026 report no undefined reference in either Paper 0 or Paper IV. Nothing else was changed by them, and no statement of either paper was added, weakened, or renumbered by these repairs; the new subsections recorded in Appendix D.6 are a separate change.

D.5 Manuscript inserts prepared from this note

The mathematical content of Sections 6 and 7 is projective: it concerns bivaluations, idempotents, and the transport of Paper IV, Definition 4.10. Paper 0's scope statement excludes "projective bivaluations, quotient towers, condensation", so that content cannot be moved into Paper 0 as it stands; the geometric fragment that Paper 0 does own — poles and pencils — is separated out.

Two self-contained inserts were therefore prepared, kept beside this document under `inserts/`:

file	target	contents
`paper-0-pole-covariance.tex`	Paper 0, new Subsection 4.4	Lemma (covariance of the pole), Proposition (rigidity along a history), Remarks on the Bruhat reading and on a prospective interface
`paper-4-transport-density.tex`	Paper IV, new Subsection 4.8	Lemma (edge criterion), Corollary (necessary condition), Propositions (log-holonomy density; normalisation against the reductive connection), Example (the $\mathbb{F}_7$ witness edge by edge), boundary remark

Both were first verified *in place* — wired into scratch copies of the host papers and built — and then integrated, in a commit of their own, by adding the two files under the respective `sections/` directories and one `\input` line each. The numbers the verification predicted are the numbers now in force: Paper 0 carries the material as Subsection 4.4 with Lemma 4.6, Proposition 4.7

and Remarks 4.8–4.9, and Paper IV as Subsection 4.8 with Lemma 4.14, Corollary 4.15, Propositions 4.16–4.17, Example 4.19 and Remark 4.20. The builds report Paper 0 at 92 pages and Paper IV at 59 pages, neither with an undefined reference.

The integration and the repairs of Appendix D.4 are deliberately two separate commits: the first adds no statement of either paper, the second adds several. Neither paper asserts any physical identification; the dictionary of Section 1 belongs to this note and is not part of the manuscripts.

D.6 What is not verified

For completeness, the following are *not* checked by the script and rest on the literature or on nothing:

(i) the identification of the $1 - \mathbf{v}_1 \cdot \mathbf{v}_2/c^2$ coefficient with the Darwin term (Warning 5.5); this is cross-checked only against the standard references [6, 3];

(ii) the claim that the Coulomb spectrum is exactly reproduced by Bohr–Sommerfeld quantization (Warning 9.7);

(iii) the geometric-quantization reading of the Reeb coordinate as a phase (Remark 2.9), which rests on the Boothby–Wang theory [11, 10] but is not instantiated here;

(iv) every statement depending on one of the five hypotheses of Section 1.4;

(v) every statement marked heuristic in Section 10.3.

References

[1] DeepSeek Harness Agent and Mingli Yuan, *Arithmetic Expression Geometry 0: Arithmetic Expression Spaces — Expansion, Motion, Ripple Duals, Commutativization, and Tearing*, manuscript, 2026. Cited as *Paper 0* with its compiled section, theorem, and equation numbers.

[2] DeepSeek Harness Agent and Mingli Yuan, *Arithmetic Expression Geometry IV: Projective Condensation and Computational Complexity*, manuscript, 2026. Cited as *Paper IV*.

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[4] L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields*, 4th ed., Course of Theoretical Physics vol. 2, Pergamon, Oxford, 1975.

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[10] H. Geiges, *An Introduction to Contact Topology*, Cambridge University Press, 2008.

[11] W. M. Boothby and H. C. Wang, "On contact manifolds", *Annals of Mathematics* **68** (1958), 721–734.